



Supplementary Materials for

GPTs are GPTs: Labor market impact potential of LLMs

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Supplementary Materials for GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models

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1 Supplementary Materials

2 Literature Review

In this section, we explore the relevant literature that this paper aims to build upon.

2.1 The Advancement of Large Language Models

The past few years have seen a surge in the development of generative AI systems, drawing considerable interest from the AI research community and the public alike. Factors such as an increase in model parameters, more training data, and advanced training architectures (1–4) have amplified their performance on a variety of complex language-based tasks. Leading large language models (LLMs) demonstrate versatility in applications such as translation, classification, creative writing, and code generation without specialized architectures or domain-specific data-tagging. These models are rapidly evolving, with their advancements readily apparent in our result analysis. Noticeable improvements are visible in Figure 1.

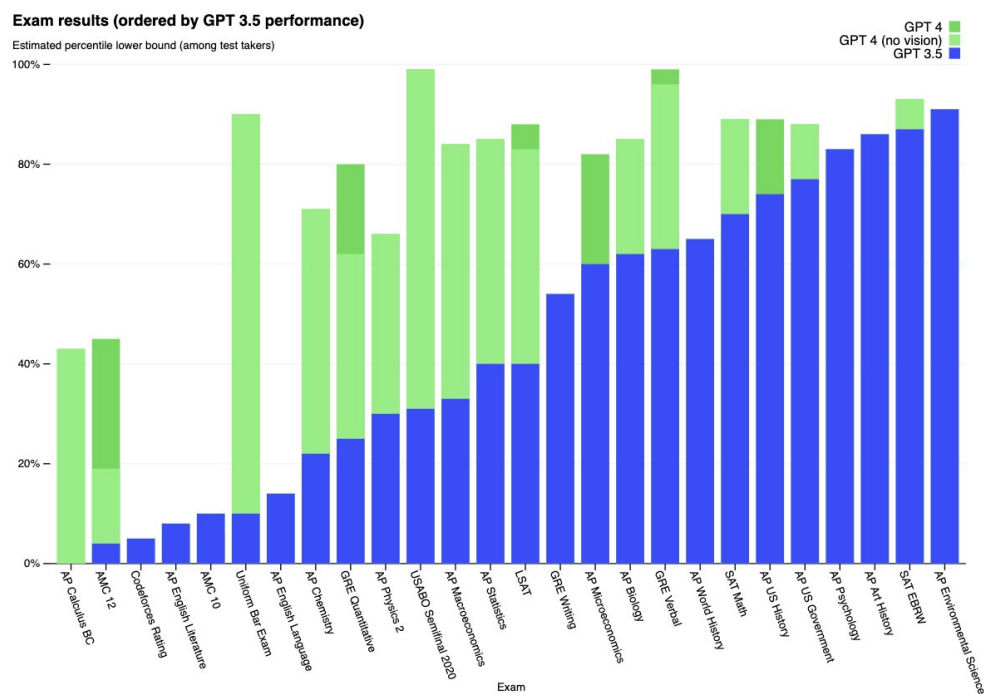


Figure 1: For a sense of how quickly model capabilities are progressing – consider the jump in exam performance between GPT-3.5 and GPT-4 (5). Figure taken from GPT-4 Technical Report (5).

Parallel advancements in enhancing the steerability and reliability of LLMs, through methods such as fine-tuning and reinforcement learning with human feedback (RLHF), can help augment a user’s perception of a model’s functionality (6, 7). Despite these improvements, many generative AI models still exhibit significant failure modes including factual inaccuracies (hallucinations or confabulations), biases, and potential security vulnerabilities (8–12). These risks pose significant constraints on the widespread adoption of generative AI systems, as many users today experience generative AI systems as task- or modality-specific modules for functions such as chatbots, image generation from captions or speech-to-text conversion.

On the other hand, this paradigm of additional training on specific rules, topics, or tasks can also be used to teach an LLM to use other digital tools such as APIs, search engines, and even other generative AI systems - a capability that can appear to multiply a model’s capabilities (13–15). Additionally, generative AI models can be combined with UI, fine-tuning (16), and retrieval corpora (17) that improve usability and performance. At the limit, these developments could usher in an era where LLMs are capable of executing any task conventionally performed at a computer.

In other words, the versatility of LLMs should be recognized as a foundational element for developing auxiliary tools. The cultivation of these tools and their integration into various systems may necessitate considerable reconfiguration of existing processes across industries and will inevitably take time. While many users might be able to use a generative AI model to write an email scheduling a meeting, domain-specific workflow integrations, encompassing specialized tooling, software, or human-supervised systems, can enable users to complete more complex tasks more efficiently.

LLMs can also be employed as valuable resources in the development of machine learning models, serving as coding assistants, data labeling services, or synthetic data generators. They could even influence economic decisions at the level of tasks, for example, by optimizing task

distribution between humans and machines (18, 19). As LLMs evolve and align more closely with user preferences, we can expect their performance to continually improve. However, it is imperative to acknowledge the serious risks accompanying these developments (20–22).

2.2 The Economic Impacts of Automation Technologies

An extensive literature addresses the labor market impacts of AI and automation technologies. The concepts of skill-biased technological change and the task model of automation – often considered the standard framework for understanding technology’s influence on labor – originated from research demonstrating that technological progress raises the demand for skilled workers over unskilled workers (23). This standard framework has seen expansions, for instance exploring automation’s effects on task-based work (24–26). This research reveals workers engaged in routine tasks to be more vulnerable to technology-induced displacement, termed routine-biased technological change. More recent research differentiates between task-displacement and task-reinstatement effects of technology, where new technology necessitates a broader spectrum of labor-intensive tasks (26, 27). The resulting wage disparity, attributed to wage reductions for routine-task workers, has been affirmed by several studies, underscoring the reshaping of the U.S. labor market by automation technologies (28–30).

Historically, automation has driven economic disparity and labor disruption, leading to negative consequences for some workers. Recent studies suggest that LLMs can provide a larger productivity boost for lower-skilled workers on identical tasks, when compared to their higher-skilled counterparts (31–33). However, many critical questions remain unanswered regarding real-world impacts on labor demand, wages, job quality, and inequality as LLMs are more widely adopted.

Previous research methods to estimate the intersection of AI capabilities and worker tasks include patent-to-task mapping (34, 35), aligning AI capabilities with occupational abilities in

O*NET (36, 37), comparing AI task benchmarks with worker tasks via cognitive abilities (38), labeling automation potential for US occupations and employing machine learning classifiers for estimation (39), task-level automation modeling and aggregation to occupational insights (40), and collecting expert forecasts (41). Pertinent to our paper, recent research has introduced new rubrics to assess the suitability of worker activities for machine learning (42, 43). These approaches have noted diversified AI technology exposure at the task-level within occupations, suggesting it is rare for any occupation to be completely subsumed by AI tools. Positive correlations have been found between automation and augmentation exposure (44), and a growing body of research investigates the specific economic impacts and opportunities of LLMs (31, 32, 37, 45–48). Our work contributes to this literature, offering insights into the potential applicability of language models within the labor market. Not all general-purpose technologies also cause automation, but sufficiently transformative advances change the composition of work.

General-purpose technologies (e.g. printing, the steam engine, electrification, computing) are characterized by widespread proliferation, continuous improvement, and the generation of complementary innovations (49, 50). Their far-reaching consequences, which unfold over decades, are difficult to anticipate, particularly in relation to labor demand (51–54). The realization of general-purpose technologies’ full potential requires extensive co-invention (49, 50, 55–57), a costly and time-consuming process involving the discovery of new business procedures (58–62). GPTs demonstrate innovational complementarities (49) as well, where GPT improvements increase productivity in downstream research and development. Consequently, many studies of machine learning technologies focus on systems-level adoption, arguing that organizational systems may require redesign to effectively take advantage of novel machine learning advancements (63–65). Appropriately designed systems can yield considerable business value and improve firm performance (66–68), with AI tools facilitating the discovery process (69–71). (65) argue that machine learning as a broad category is likely a general-purpose technology.

Our evidence supports a wider impact, as even subsets of machine learning software meet the criteria for general-purpose technology status independently. This paper provides a set of measurements of LLM impact potential and demonstrates the use case of applying LLMs to develop such measurements efficiently and at scale. In doing so, we showcase the general-purpose potential of LLMs. The primary contribution of the paper is to motivate consideration of the difference in potential impact with and without additional software tools that complement LLMs. This helps to evaluate LLMs as general-purpose technologies, particularly in light of the increased exposure of R&D-related occupations to these systems. By employing task-level information to assess whether LLMs fulfill the criteria of a general-purpose technology, we seek to merge the two perspectives for understanding the technology-labor relationship.

We attempt to build on these diverse literature streams in several ways. Echoing (37), we focus our analysis on the impact of LLMs, rather than addressing machine learning or automation technologies more broadly. LLMs as a category include a few types of machine learning methods, and do have automation potential. Additionally, we propose a novel method that employs LLMs, specifically GPT-4, to assess tasks for exposure and automation potential, thereby bolstering human scoring efforts.¹ Subsequently, we aggregate our findings to occupations and industries, capturing the overall potential exposure in the contemporary U.S. labor market.

While task impact potential is extensive, integration of LLMs and LLM-powered software into broader systems is crucial for maximizing potential. The diffusion of such general-purpose technologies into economic applications may initially encounter co-invention barriers since predicting the need for human oversight remains challenging, especially for tasks where model capabilities equal or exceed human proficiency. Although initial human supervision norms and requirements may decelerate technology diffusion, users are likely to become increasingly

¹These techniques have been applied in a number of papers since this paper’s first release. We discuss many of the trade-offs and limitations of this approach, especially that GPT-4 ratings are sensitive to prompt design and subject to a number of limitations. See Section 3.3.

familiar with LLMs and LLM-powered systems over time, particularly in discerning when and how to trust their outputs. Systems may also become more capable over time. It's conceivable that for many tasks, the value of time savings and seamless integration will outweigh that of quality improvements. It's also possible that the initial focus of development will be on augmentation, followed by automation (72).

General-purpose technologies are relatively rare. They are characterized by their pervasiveness, improvement over time, and the development of significant co-invention and spillovers (50). Our tests to determine if "GPTs are GPTs" are primarily tests of exposure pervasiveness and whether complementary innovations are needed to unlock otherwise feasible LLM applications. We find evidence that both of these conditions are likely.

If "GPTs are GPTs," the eventual trajectory of LLM development and application may be challenging for policymakers to predict and regulate. As with other general-purpose technologies, much of these models' potential will emerge across a broad range of economically valuable use cases, including the creation of new types of work (26, 44), capital and labor augmentation, and automation. Our research attempts to measure what is technically feasible now and what may be feasible with future complementary technology development, but by default our approach has a limited ability to predict the evolving impact potential of the LLMs over time.

3 Rubrics

In our discussion, we use the terms "LLMs" and "GPTs" interchangeably to refer specifically to the GPT-family of models, including those accessible via platforms such as ChatGPT or the OpenAI Playground. This included models like GPT-3.5 and GPT-4 at the time of assessment. Our focus is primarily on LLMs that can generate text and code.

The broader term "generative AI" encompasses other types of models capable of processing and producing various data types, such as images or audio. "LLM-powered software" refers to

applications built upon LLMs or those that integrate LLMs with other generative AI models.

3.1 Overall Exposure

3.1.1 Overall Exposure Rubric Summary

Summary of exposure rubric

No exposure (E0) if:

- using the described LLM results in no or minimal reduction in the time required to complete the activity or task while maintaining equivalent quality^a or
- using the described LLM results in a decrease in the quality of the activity/task output.

Direct exposure (E1) if:

- using the described LLM via ChatGPT or the OpenAI playground can decrease the time required to complete the DWA or task by at least half (50%).

LLM+ Exposed (E2) if:

- access to the described LLM alone would not reduce the time required to complete the activity/task by at least half, but
- additional software could be developed on top of the LLM that could reduce the time it takes to complete the specific activity/task with quality by at least half. Among these systems, we count access to image generation systems.^b

^aEquivalent quality means that a third party, typically the recipient of the output, would not notice or care about LLM assistance.

^bIn practice, we categorize access to image capabilities separately (E3) to facilitate annotation, though we combine E2 and E3 for all analyses.

3.1.2 Full Exposure Rubric

E Exposure Taxonomy

Consider the most powerful OpenAI large language model (LLM) This model can complete many tasks that can be formulated as having text input and text output where the context for the input can be captured in 2000 words. The model also cannot draw up-to-date facts (those from <1 year ago) unless they are captured in the input.

Assume you are a worker with an average level of expertise in your role trying to complete the given task. You have access to the LLM as well as any other existing software or computer hardware tools mentioned in the task. You also have access to any commonly available technical tools accessible via a laptop (e.g. a microphone, speakers, etc.). You do not have access to any other physical tools or materials.

Please label the given task according to the taxonomy below.

E0 — No exposure

Label tasks E0 if direct access to the LLM through an interface like ChatGPT or the OpenAI playground cannot reduce the time it takes to complete this task with equivalent quality by half or more.

206
207 If a task requires a high degree of human interaction (for example, in person demonstrations) then it should be classified as
208 E0.
209
210 ## E1 — Direct exposure
211
212 Label tasks E1 if direct access to the LLM through an interface like ChatGPT or the OpenAI playground alone can reduce the
213 time it takes to complete the task with equivalent quality by at least half. This includes tasks that can be reduced to:
214 — Writing and transforming text and code according to complex instructions,
215 — Providing edits to existing text or code following specifications,
216 — Writing code that can help perform a task that used to be done by hand,
217 — Translating text between languages,
218 — Summarizing medium-length documents,
219 — Providing feedback on documents,
220 — Answering questions about a document, or
221 — Generating questions a user might want to ask about a document.
222
223 ## E2 — Exposure by LLM-powered applications
224
225 Label tasks E2 if having access to the LLM alone may not reduce the time it takes to complete the task by at least half, but
226 it is easy to imagine additional software that could be developed on top of the LLM that would reduce the time it takes
227 to complete the task by half. This software may include capabilities such as:
228 — Summarizing documents longer than 2000 words and answering questions about those documents
229 — Retrieving up-to-date facts from the Internet and using those facts in combination with the LLM capabilities
230 — Searching over an organization’s existing knowledge, data, or documents and retrieving information
231
232 Examples of software built on top of the LLM that may help complete worker activities include:
233 — Software built for a home goods company that quickly processes and summarizes their up-to-date internal data in customized
234 ways to inform product or marketing decisions
235 — Software that is able to suggest live responses for customer service agents speaking to customers in their company’s
236 customer service interface
237 — Software built for legal purposes that can quickly aggregate and summarize all previous cases in a particular legal area and
238 write legal research memos tailored to the law firm’s needs
239 — Software specifically designed for teachers that allows them to input a grading rubric and upload the text files of all
240 student essays and have the software output a letter grade for each essay
241 — Software that retrieves up-to-date facts from the internet and uses the capabilities of the LLM to output news summaries in
242 different languages
243
244 ## E3 — Exposure given image capabilities
245
246 Suppose you had access to both the LLM and a system that could view, caption, and create images. This system cannot take video
247 media as inputs. This system cannot accurately retrieve very detailed information from image inputs, such as
248 measurements of dimensions within an image. Label tasks as E3 if there is a significant reduction in the time it takes
249 to complete the task given access to a LLM and these image capabilities:
250 — Reading text from PDFs,
251 — Scanning images, or
252 — Creating or editing digital images according to instructions.
253
254 ## Annotation examples:
255
256 Occupation: Inspectors, Testers, Sorters, Samplers, and Weighers
257 Task: Adjust, clean, or repair products or processing equipment to correct defects found during inspections.
258 Label (E0/E1/E2/E3): E0
259 Explanation: The model does not have access to any kind of physicality, and more than half of the task (adjusting, cleaning
260 and repairing equipment) described requires hands or other embodiment.
261
262 Occupation: Computer and Information Research Scientists
263 Task: Apply theoretical expertise and innovation to create or apply new technology, such as adapting principles for applying
264 computers to new uses.
265 Label (E0/E1/E2/E3): E1
266 Explanation: The model can learn theoretical expertise during training as part of its general knowledge base, and the
267 principles to adapt can be captured in the text input to the model.
268
269 Activity: Schedule dining reservations.
270 Label (E0/E1/E2/E3): E2
271 Explanation: Automation technology already exists for this (e.g. Resy) and it’s unclear what an LLM offers on top of using
272 that technology (no-diff). That said, you could build something that allows you to ask the LLM to make a reservation on
273 Resy for you.
274
275 Activity: Negotiate purchases or contracts.
276 Label (E0/E1/E2/E3): E2
277 Explanation: You could have each party transcribe their point of view and then feed this to an LLM to resolve any disputes (E3
278). That said, many people would need to buy into using new technological tools to accomplish this (system).
279
280 Occupation: Allergists and Immunologists
281 Task: Prescribe medication such as antihistamines, antibiotics, and nasal, oral, topical, or inhaled glucocorticosteroids.
282 Label (E0/E1/E2/E3): E2
283 Explanation: The model can provide guesses for different diagnoses and write prescriptions and case notes. However, it still
284 requires a human in the loop using their judgment and knowledge to make the final decision.
285
286

3.1.3 Key Additional Exposure Rubric Results

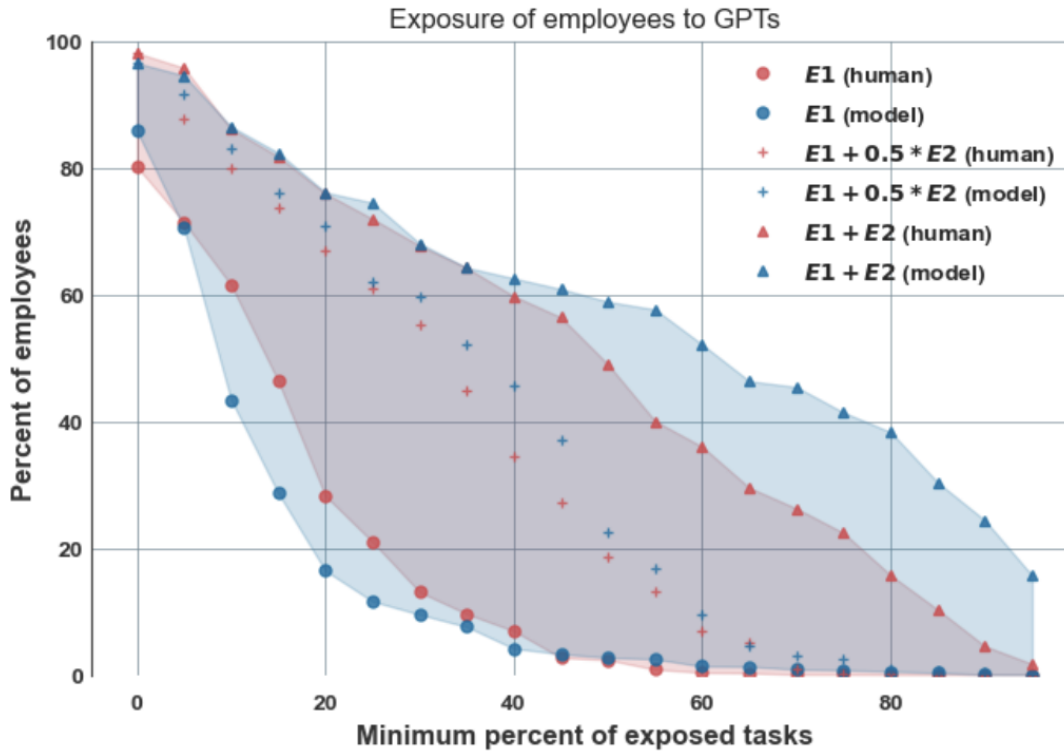


Figure 2: Worker exposure across the economy, at different levels of exposure ($E1$, $E1+0.5 \cdot E2$, $E1+E2$). Approximately 80 percent of workers have at least 10 percent of their tasks exposed at the $E1+0.5 \cdot E2$ level. 18.5 percent of workers have more than 50 percent of their tasks exposed at the same level.

3.2 Automation Exposure

We implemented an automation-focused rubric as well, though we only retrieved labels using GPT-4 and did not validate these results against labels provided by humans.

3.2.1 Automation Rubric Prompt

```
# T Automation Rubric
1. Determine if the occupation/task pair meets the definition of T0 No-Automation Exposure. If it does, label it as T0 and stop.
2. If the occupation/task pair does not meet the definition of T0 No-Automation Exposure, determine if the occupation/task pair meets one of the other definitions above and select the label that applies:
- T4: Full automation exposure
- T3: High automation exposure
```

302 — T2: Moderate automation exposure
 303 — T1: Low automation exposure
 304
 305 ## Rubric
 306
 307 LLM++ is a software tool that can perform a wide range of digital tasks but cannot program physical machines or objects.
 308
 309 Consider the most powerful OpenAI large language model (LLM). This model can complete many tasks that can be formulated as
 310 having text, image, or audio input and text, image, or audio output. The model also cannot draw up-to-date facts (those
 311 from <1 year ago) unless they are captured in the input. This system cannot accurately retrieve very detailed
 312 information from image inputs, such as measurements of dimensions within an image.
 313
 314 On its own, the LLM can perform tasks that can be reduced to:
 315 — Writing and transforming text and code according to complex instructions,
 316 — Providing edits to existing text or code following specifications,
 317 — Writing code that can help perform simple, repetitive digital tasks,
 318 — Debugging code or software
 319 — Programming in computer languages like Python, C++, Java, Stata
 320 — Assisting with a data analysis
 321 — Translating text between languages,
 322 — Summarizing short documents,
 323 — Providing feedback on documents,
 324 — Answering questions about a document,
 325 — Generating questions a user might want to ask about a document,
 326 — Writing questions for an interview or assessment,
 327 — Writing and responding to emails, including ones that involve refuting information or engaging in a negotiation (but only if
 328 the negotiation is via written correspondence),
 329 — Maintaining records of written data,
 330 — Prepare training materials based on general knowledge,
 331 — Searching for a relevant document over a small number of documents,
 332 — Reading text from PDFs,
 333 — Scanning images,
 334 — Creating or editing digital images according to instructions,
 335 — Captioning an image,
 336 — Transcribing text, or
 337 — Informing anyone of any information via any written or spoken medium.
 338
 339 Assume you are a worker with an average level of expertise in your role trying to complete the given task. You have access to
 340 the LLM as well as any other existing software or computer hardware tools mentioned in the task. You also have access to
 341 any commonly available technical tools accessible via a laptop (e.g. a microphone, speakers, etc.). You do not have
 342 access to any other physical tools or materials.
 343
 344 You also have access to LLM++ which is additional software developed on top of the LLM. LLM++ includes capabilities such as:
 345 — Summarizing documents longer than 2000 words and answering questions about those documents,
 346 — Retrieving up-to-date facts from the Internet and using those facts in combination with the LLM capabilities,
 347 — Searching over an organization's existing knowledge, data, or documents and retrieving information,
 348 — Retrieving highly specialized domain knowledge,
 349 — Making recommendations given data or written input,
 350 — Analyzing written information to inform decisions,
 351 — Preparing training materials based on highly specialized knowledge,
 352 — Performing legal research,
 353 — Providing counsel on issues,
 354 — Writing lesson plans, and
 355 — Maintaining databases.
 356
 357 We refer to this suite of software tools as LLM++. LLM++ can NOT program physical machines or objects.
 358
 359 Please label the given task according to the rubric below.
 360
 361 ****T0 No—Automation Exposure**** A class of tasks for which LLM++ can not conceivably perform any aspect of the task in any
 362 manner.
 363
 364 ****T4 Full Automation Exposure**** A class of tasks where, in most contexts in which this task is currently performed by a human,
 365 LLM++ can complete all aspects of this task with high quality when prompted by a human. The output does not normally
 366 require oversight by a human. Oversight is not normally required for tasks labeled T4 because the consequences for
 367 failure or inaccuracy are small for this task, human judgment is not necessary to complete this task, and generative
 368 models can consistently perform this task with very high quality.
 369
 370 ****T3 High Automation Exposure**** A class of tasks where, in most contexts in which this task is currently performed by a human,
 371 LLM++ could complete 90–100% of the components of the task when prompted, but the output requires oversight from a
 372 human. Oversight is normally required because the consequences for failure or inaccuracy are significant for this task,
 373 human judgment is necessary to complete this task, and/or generative models cannot perform all aspects of this task with
 374 high quality consistently. These tasks rely almost exclusively on the processing of digital information, but human
 375 judgment is needed to ensure that any digital outputs from LLM++ are high enough quality to be acceptable for the
 376 particular context.
 377
 378 ****T2 Moderate Automation Exposure**** A class of tasks where, in most contexts in which this task is performed by a human, LLM
 379 ++ could complete between 50%–90% of the components of the task at high quality. These tasks normally rely heavily on
 380 the processing of digital information, but a significant portion of the task also involves actions that LLM++ cannot
 381 perform with high quality. These tasks require at least some human action beyond just double-checking generative model
 382 outputs (such as interpretation, judgment, human-to-human communication, or physical actions).
 383
 384 ****T1 Low Automation Exposure**** A class of tasks where, in most contexts in which this task is performed by a human, LLM++
 385 could complete between 0%–50% of the components of the task at high quality. These tasks normally rely only partially

on the processing of digital information, while the majority of the task involves actions that LLM++ cannot perform with high quality. A majority of the actions that need to be taken to complete this task require a human to perform the action.

Definitions

High quality means someone receiving or reviewing the output would not be able to tell the difference between whether it came from LLM++ or a human. For tasks that require a lot of interaction during the completion of the task (e.g. meetings, negotiations), high quality means the people you were interacting with either would not know or would not care that they were interacting with LLM++.

Digital information or **information that can easily be expressed digitally** includes but is not limited to text, audio, images, PDFs, books, and code.

Annotation Examples

Occupation: Special Education Teachers, Preschool
Task: Develop individual educational plans (IEPs) designed to promote students' educational, physical, or social development.
Automation score (T0/T1/T2/T3/T4): T2
Explanation: While LLM++ can assist in the creation of educational plans, this task requires significant interpretation and judgment by a human in order to be effective. Special Education Teachers must consider the individual needs and abilities of each student, as well as communicate with parents and other educators to ensure the plan is appropriate and effective. While LLM++ can provide suggestions and potential plans, it cannot make the final decision or communicate with stakeholders.

Occupation: Construction and Building Inspectors
Task: Inspect and monitor construction sites to ensure adherence to safety standards, building codes, or specifications.
Automation score (T0/T1/T2/T3/T4): T1
Explanation: While LLM++ can provide some support in the inspection and monitoring of construction sites, this task involves a significant amount of physical inspection and communication with workers and supervisors on-site. Additionally, the inspector may need to make on-the-spot judgments based on what they observe and may need to refer to local building codes and regulations which may not be available in digital format. As such, this task requires a significant amount of human input and oversight.

Occupation: Food and Tobacco Roasting, Baking, and Drying Machine Operators and Tenders
Task: Dump sugar dust from collectors into melting tanks and add water to reclaim sugar lost during processing.
Automation score (T0/T1/T2/T3/T4): T0
Explanation: This task requires physical actions that cannot be completed by LLM++.

Occupation: Historians
Task: Translate or request translation of reference materials.
Automation score (T0/T1/T2/T3/T4): T4
Explanation: LLM++ can easily complete this task by translating text from one language to another. Human intervention is not usually necessary.

Occupation: Electronics Engineer
Task: Prepare documentation containing information such as confidential descriptions or specifications of proprietary hardware or software, product development or introduction schedules, product costs, or information about product performance weaknesses.
Automation score (T0/T1/T2/T3/T4): T2
Explanation: LLM++ can assist with documentation but human judgment is needed for understanding specifications, costs, and performance weaknesses. Thus, this task falls under T2 Moderate Automation Exposure.

Occupation: Advanced Practice Psychiatric Nurse
Task: Write prescriptions for psychotropic medications as allowed by state regulations and collaborative practice agreements.
Automation score (T0/T1/T2/T3/T4): T1
Explanation: LLM++ could be used to help create prescriptions given digital input about the patient's context, but a human would need to double-check the information to ensure that it meets regulations and is appropriate for the patient.

Now, apply the above rubric to the example below:

3.2.2 Expanded Discussion of the Automation Rubric Results

Our automation exposure measures are a proof-of-concept that GPT-4 may be a useful tool for researchers looking to label tasks with a text-based rubric, but we have not validated the GPT-4 evaluations against the judgement of human raters. Furthermore, this rubric has many of the same weaknesses as the exposure rubric. The results we find, however, suggest some intuitive findings and broadly have high correlation with the exposure measures. Whether this is because

GPT-4 as a labeler cannot tease out the nuanced differences between the rubrics or because automation and augmentation are actually highly correlated is unclear. These are still early results. To convert our automation labels from the rubric into numerical scores between 0 and 1, we use the following mapping:

- T0 (no exposure to automation) is assigned a score of 0.
- T1 is assigned a score of 0.25.
- T2 is assigned a score of 0.5.
- T3 is assigned a score of 0.75.
- T4 (full exposure to automation) is assigned a score of 1.

Preliminary results suggest exposure as defined in the E rubric scores from GPT-4 and automation as defined in the T rubric have a strong positive association. Table 1 contains summary statistics on our GPT-4 evaluations of the automation rubric. Table 2 shows the regression results from a regression of GPT-4 $E1 + 0.5 * E2$ exposure scores on GPT-4 automation ratings. Figure 3 shows the scatterplot of the same exposure scores against the automation scores we generate at the task level on top, and the bottom panel of Figure 3 shows the binscatter of automation score exposure against log wage rated at the occupation-level. We see similar trends in these early GPT-4 measures of automation as we do in overall exposure.

Overall exposure and automation exposure are highly correlated, with an R^2 of 55.6 percent in a univariate regression of exposure scores on automation scores at the task-level. This means that where automation is high, it is likely the case that exposure, which includes augmentation potential, is also relatively high.

Automation exposure ratings follow a similar trend to general exposure ratings (see Figure 3). Higher wage occupations tend to be more likely to have tasks with more automatable components.

No occupation has 100 percent of tasks exposed to automation by GPT under this rubric, though many occupations have high levels of exposure that suggest they might be reconfigured entirely. Occupations with the highest automation exposure include telephone operators, travel agents, word processors and typists, credit authorizers, checkers, and clerks, desktop publishers, and telemarketers. Overall, only 1.86 percent of tasks are assessed to be exposed to “full automation” suggesting that there may be significant last-mile barriers to automating many tasks. Whereas there are a number of knowledge-intensive roles at the top of the list of most exposed occupations, the top automation-exposed occupations tend to include more routine cognitive work.

In light of this, emphasizing augmentation instead of automation may not be useful in charting a path forward for designing jobs, as it may be that any model capable of having a meaningful augmentation impact will also carry strong automation risk. Even in cases where it is possible to prioritize augmentation, market-level factors like price and income elasticities of supply and demand (among others) dictate the extent to which individual workers are aided or displaced by new configurations of work (73). Whether workers are displaced or complemented on balance will depend on the particulars of the application and emergent use cases as new systems and complements emerge. Based on this evidence, it is also premature to suggest that high broad-based exposure implies the wholesale elimination of jobs for similar reasons.

Table 1: Distribution of GPT-4 Automation Annotations for LLM++ Across Tasks

Automation	Task Count	Task Share
No automation exposure	5,528	28.69%
Low automation exposure	2,937	15.25%
Moderate automation exposure	7,796	40.47%
High automation exposure	2,646	13.73%
Full automation exposure	358	1.86%
Total	19,265	100%

Column 1 lists the five possible GPT-4-generated annotations on the automation taxonomy for LLM++. It is important to remember that these annotations are generated based on the definition given of LLM++ in the taxonomy, which includes not only LLM capabilities, but the potential for software built on top of an LLM to automate the task. The number of tasks assigned to each label is presented in column 2, and the share of all tasks assigned the label is presented in column 3.

	GPT-4 Exposure Rating 1	GPT-4 Exposure Rating 2
	(1)	(2)
GPT-4 Automation Score	0.936*** (0.00609)	0.847*** (0.00737)
Constant	0.045*** (0.00201)	-0.005** (0.00209)
N	19265	19265
R^2	0.556	0.457

Table 2: Regression of GPT-4 Exposure Rubric Scores on GPT-4 Automation Rubric Scores

Table Notes: The table shows the regression of our GPT-4 Exposure Rubric scores following the $E1 + 0.5 * E2$ convention described above on the GPT-4 Automation Score as defined by the Automation (“T”) rubric. The unit of observation is the task. T0, T1, T2, T3, and T4 are mapped to scores of 0, 0.25, 0.5, 0.75, and 1 respectively. In both cases, the automation score explains about half of the variance of the exposure scores.

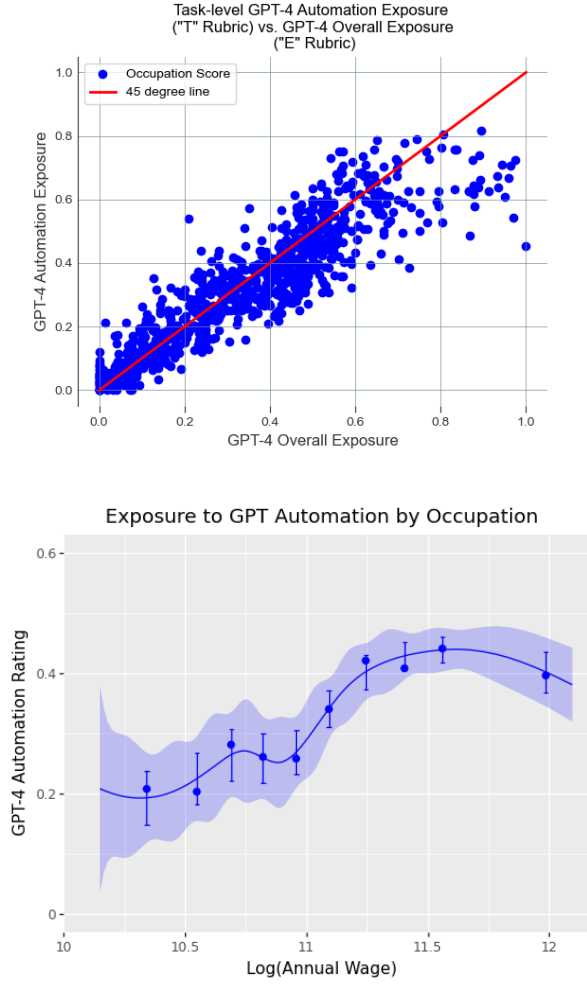


Figure 3: GPT-4 Ratings of Automation Exposure (“T”) vs. Overall Exposure (“E”) (top) and Binscatter of GPT-4 Automation Exposure Ratings by Occupation vs. Log Wages (bottom)

3.3 Limitations of our methodology

At this stage, some general-purpose technology criteria are easier to evaluate than others. Our primary focus at this early stage is to test the hypothesis that LLMs have a pervasive influence on the economy, similar to the approach taken by (65), who analyzed machine learning diffusion through job postings to assess its status as a general-purpose technology. Instead of using job postings or studying machine learning in general, we employ the task evaluation approach with both human and GPT-4 annotations. This analysis may reveal whether the impacts are limited to

a specific set of similar tasks or occupations or if they will be more widespread.

3.3.1 Subjective human judgments

A fundamental limitation of our approach lies in the subjectivity of the labeling. In our study, we employ six annotators who are familiar with LLM capabilities and have extensive experience evaluating LLM capabilities and outputs. However, this group is not occupationally diverse, potentially leading to biased judgments regarding LLMs’ reliability and effectiveness in performing tasks within unfamiliar occupations. Furthermore, annotators are not experts on complementary software needed to leverage LLMs. In an effort to reduce subjectivity, the exposure rubric provided detailed examples of the kinds of software tools commonly built around base LLMs at the time of task-labeling, including integrations with web browsers or with databases with internal company data (see 3.1.2 for the full rubric and set of annotation instructions with examples. We acknowledge that obtaining high-quality labels for each task in an occupation requires workers engaged in those occupations or, at a minimum, possessing in-depth knowledge of the diverse tasks within those occupations. This represents an important area for future work in validating these results.

3.3.2 Measuring LLMs with GPT-4

Recent research indicates that GPT-4 serves as an effective discriminator, capable of applying intricate taxonomies and responding to changes in wording and emphasis (5). The outcomes of GPT-4 task classification are sensitive to alterations in the rubric’s wording, the prompt’s order and composition, the presence or absence of specific examples in the rubric, the level of detail provided, and the definitions given for key terms. Iterating on the prompt, based on observed outcomes in a small validation set, can enhance the agreement between model outputs and the rubric’s intent. Consequently, there are slight differences between the rubric presented to humans and the one used for GPT-4. This decision was made deliberately to guide the model towards

reasonable labels without excessively influencing human annotators. As a result, we use multiple annotation sources, but none should be considered the definitive ground truth relative to the others. In this analysis, we present results from human annotators as our primary results. Further improvement and innovation in crafting effective rubrics for LLM classification remains possible. Still, we observe a high degree of agreement between human ratings and GPT-4 ratings at the occupation level concerning overall exposure to LLM systems (see Figure 4). At the task level, GPT-4 and human raters agreed 66 percent of the time.

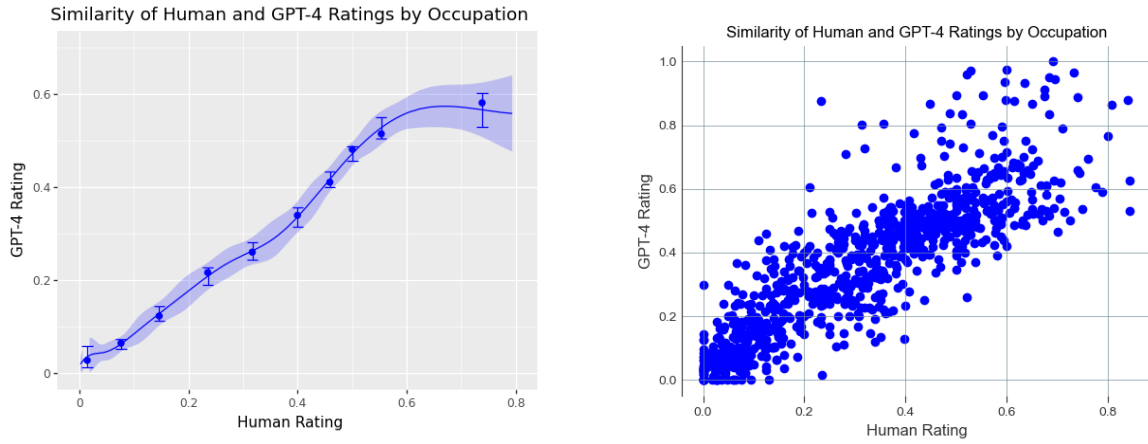


Figure 4: Human raters (x-axis) and GPT-4 ratings (y-axis) show a high degree of agreement about LLM exposure by occupation. Near the highest levels of exposure following the $E1+0.5 \cdot E2$ method of aggregating exposure scores to occupations, GPT-4 ratings tend to be lower than Human ratings.

3.3.3 Additional Weaknesses

- **Validity of task-based framework.** It is unclear to what extent occupations can be entirely broken down into tasks, and whether this approach systematically omits certain categories of skills or tasks that are tacitly required for competent performance of a job. Additionally, tasks can be composed of sub-tasks, some of which are more exposed than others. Some tasks may function as precursor to other tasks, such that the completion of downstream tasks

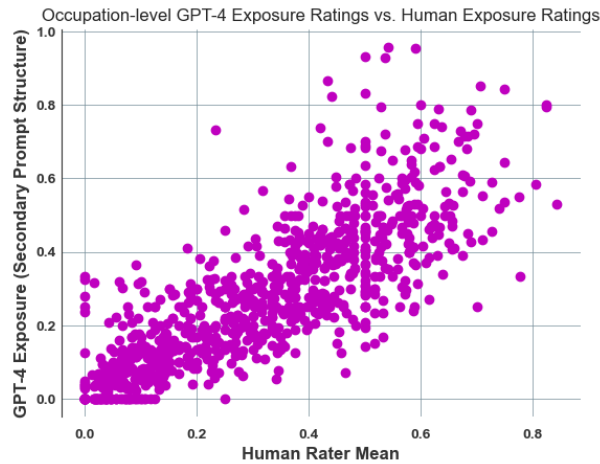


Figure 5: As in the above figure, human ratings are on the x-axis at the occupation-level, but plotted against a secondary GPT-4 prompt on the y-axis. The results are qualitatively similar under both prompts.

is dependent on precursor tasks. If the task-based breakdown is not a valid representation of how most work in an occupation is performed, our occupation-level exposure analysis would be significantly less informative.

- **Lack of expertise and task interpretation.** Human annotators were mostly unaware of the specific occupations mapped to each task during the labeling process. This led to unclear logic for aggregating tasks and occupations. Ultimately, we collected additional labels for task/occupation pairs where there was significant disagreement.

- **Forward-looking and subject to change, with some early evidence.** Accurately predicting future LLM applications remains a significant challenge, even for experts. The discovery of new emergent capabilities, changes in human perception biases, and shifts in technological development can all affect the accuracy and reliability of predictions regarding the potential impact of LLMs on worker tasks and the development of LLM-powered software. Our projections are inherently forward-looking and based on current trends, evidence, and perceptions of technological possibilities. As a result, they may change with new advances

550 in the field. For example, some tasks that seem unlikely for LLMs or LLM-powered
551 software to impact today might change with the introduction of new model capabilities.
552 Conversely, tasks that appear exposed might face unforeseen challenges limiting language
553 model applications.

554 Future research should consider two additional studies: one exploring LLM adoption
555 patterns across sectors and occupations, and another scrutinizing the actual capabilities and
556 limitations of state-of-the-art models in relation to worker activities beyond the scope of our
557 exposure scores. For example, despite GPT-4's recent advances in multimodal capabilities,
558 our E1 ratings on direct LLM exposure did not account for vision capabilities. Future work
559 should consider the impact of such advancing capabilities. Moreover, we acknowledge
560 potential discrepancies between theoretical and practical performance, particularly in
561 complex, open-ended, and domain-specific tasks.

- 562 • **Sources of disagreement.** While we did not rigorously examine sources of disagreement,
563 we found a few places where humans and the model tended to get "stuck" in their
564 assessments:

- 565 – Tasks or activities where while an LLM could theoretically help or accomplish the
566 task, adopting it to do so would require multiple people to change their habits or
567 expectations (e.g. meetings, negotiations),
- 568 – Tasks or activities where there is currently some regulation or norm that requires or
569 suggests human oversight, judgment or empathy (e.g. making decisions, counseling),
570 and
- 571 – Tasks or activities where there already exists a technology that can reasonably
572 automate the task (e.g. making reservations).

573 • **Other barriers to deployment.** The functionality of large language models (LLMs) is
574 significantly influenced by the degree of trust users accord them and the users' adaptability.
575 For example, in the legal realm, the utility of models is contingent upon the professionals'
576 trust in model outputs without recourse to corroborate original documents or conduct
577 independent research. Other factors such as technology affordability, adaptability, worker
578 and firm predilections, and incentives critically shape the assimilation of LLM-based tools.
579 Consequently, mitigating ethical and safety risks associated with LLMs, including bias,
580 misinformation, and misalignment, could boost their acceptance (11). The adoption of
581 LLMs, however, will vary across economic sectors due to aspects like data accessibility,
582 regulatory landscape, and power-interest distribution. Hence, gaining comprehensive
583 insights into LLM adoption and usage by workers and firms necessitates a deeper dissection
584 of these subtleties, which our rubrics currently overlook.

585 • **Possible assumption of augmentation given rubric design.** Our principal findings
586 derive from a prompt assessing potential for time reduction in task completion without
587 compromise on quality, which might inadvertently assume task augmentation as the primary
588 exposure vector. While we refrain from arbitrating between augmentation, automation,
589 or other technological impacts, we introduce an automation rubric with a distinct prompt
590 aimed at measuring the replacement potential of human labor by machines. Our results
591 generally align with the variation in exposure scores, partially attributed to automation
592 exposure. A very low fraction of all tasks were assessed to be fully automatable (approx.
593 1.3 percent), but those tasks with high levels of exposure were assessed to have a larger
594 share of task-components that could plausibly be automated. However, it's noteworthy that
595 our automation scores have been annotated exclusively by GPT-4, without human evaluator
596 involvement. They should be interpreted with caution.

- **Software as an arbitrary criterion for co-invention.** We discuss the expansion of task exposure when additional software tools are built around LLMs. This is a narrow and arbitrary addition to the baseline capabilities of LLMs. Evaluators consider what might be possible with respect to additional software. Imagining additional software capabilities is inherently difficult and should be expected to lead to imprecise exposure estimates. Furthermore, software complements are only a narrow slice of what will be necessary to unlock gains from generative AI (49). Necessary changes also include business process innovation, worker training, and in some cases transformational changes in production.
- **Exclusion of physical work.** We do not include possible physical applications of LLMs, including their use in robotics, when we consider what tasks may be exposed to LLMs.

4 Data Collection

4.1 Data on Activities and Tasks Performed by Occupation in the U.S. (also in main text)

We use the O*NET 27.2 database (74), which contains information on 1,016 occupations, including their respective Detailed Work Activities (DWAs) and tasks. A DWA is a comprehensive action that is part of completing task, such as "Study scripts to determine project requirements." A task, on the other hand, is an occupation-specific unit of work that may be associated with zero, one, or multiple DWAs. We offer a sample of tasks and DWAs in Table 3. The two datasets we use consist of:

- 19,265 tasks, consisting of a "task description" and a corresponding occupation, and
- 2,087 DWAs, where most DWAs are connected to one or more tasks, and tasks may be associated with one or more DWAs, though some tasks lack any associated DWAs.

Task ID	Occupation Title	DWAs	Task Description
14675	Computer Systems Engineers/Architects	Monitor computer system performance to ensure proper operation.	Monitor system operation to detect potential problems.
18310	Acute Care Nurses	Operate diagnostic or therapeutic medical instruments or equipment. Prepare medical supplies or equipment for use.	Set up, operate, or monitor invasive equipment and devices, such as colostomy or tracheotomy equipment, mechanical ventilators, catheters, gastrointestinal tubes, and central lines.
4668.0	Gambling Cage Workers	Execute sales or other financial transactions.	Cash checks and process credit card advances for patrons.
15709	Online Merchants	Execute sales or other financial transactions.	Deliver e-mail confirmation of completed transactions and shipment.
6529	Kindergarten Teachers, Except Special Education	–	Involve parent volunteers and older students in children’s activities to facilitate involvement in focused, complex play.
6568	Elementary School Teachers, Except Special Education	–	Involve parent volunteers and older students in children’s activities to facilitate involvement in focused, complex play.

Table 3: Sample of occupations, tasks, and Detailed Work Activities from the O*NET database. We see that aggregating over activities alone is imprecise, as evidenced by the fact that we’d expect Gambling Cage Workers to complete the given DWA in person, using some physicality while we’d expect Online Merchants to complete the same activity solely with a computer.

4.2 Agreement Statistics and Figures

We also include a confusion matrix for the human ratings vs. GPT-4 exposure rubric. The cells show the counts of tasks given each label by the labeled approach represented on each axis.

4.3 Data on Wages, Employment, and Demographics

We obtain employment and wage data from the 2020, 2021, and 2022 Occupational Employment series provided by the Bureau of Labor Statistics. This dataset encompasses occupational titles, the number of workers in each occupation, and occupation-level employment projections for 2032, typical education required for entry in an occupation and on-the-job training required to attain competency in an occupation (75). We use the BLS-recommended crosswalk to O*NET (76) to link the O*NET task and DWA dataset and the BLS Labor Force Demographics (77), which is derived from the Current Population Survey (CPS). Both of these data sources are collected by the U.S. government and primarily capture workers who are not self-employed, are documented,

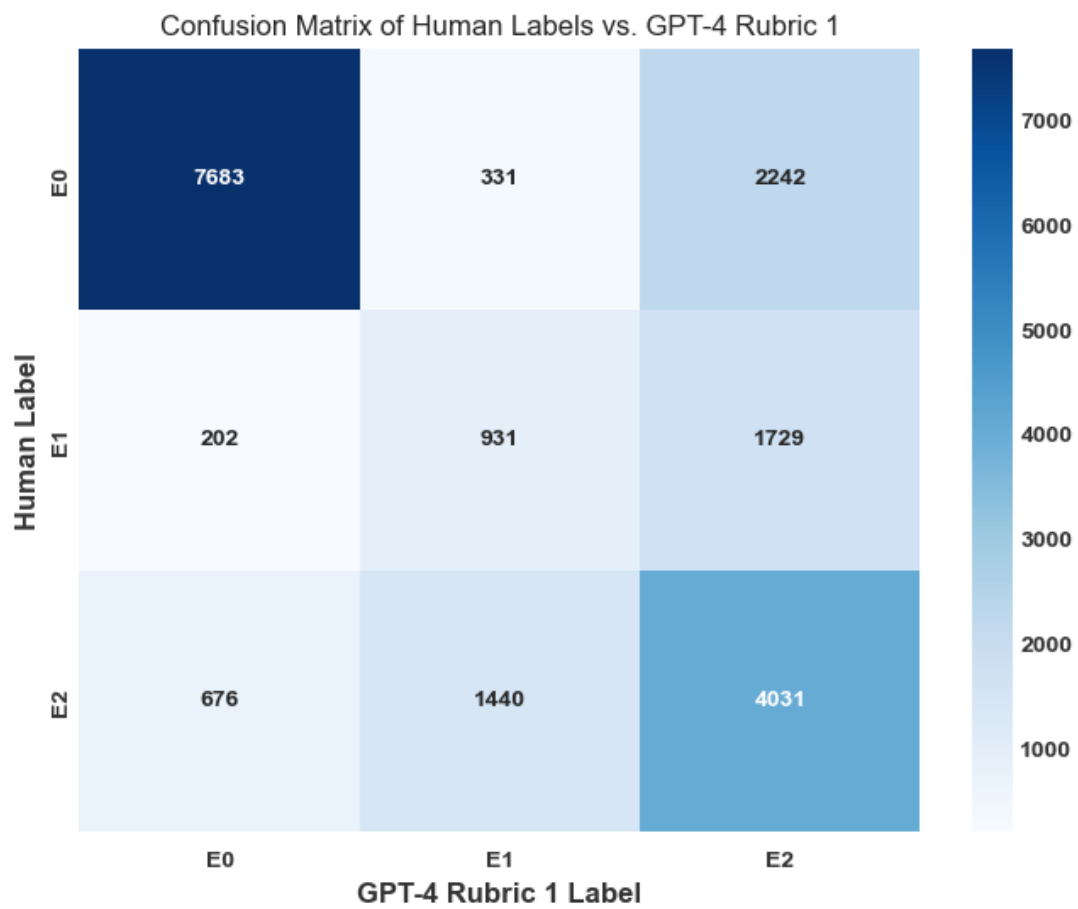


Figure 6: Human vs. Main GPT-4 Rubric Evaluations by Task

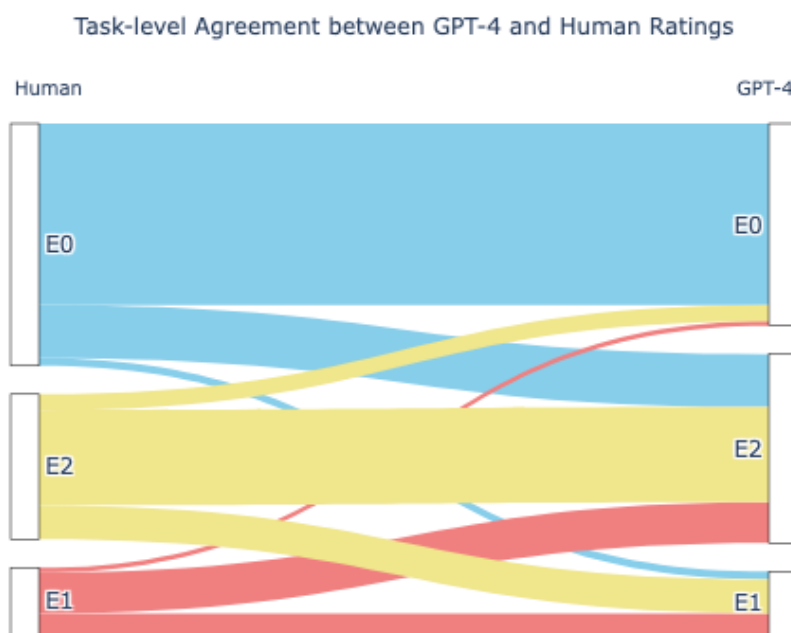


Figure 7: Sankey Diagram showing task level agreement flows between GPT-4 ratings and Human ratings. Overall, GPT-4 agreed with human raters on 66 percent of tasks.

and are working in the so-called formal economy.²

5 Skills

5.1 Skill Importance

In this section, we explore the relationship between skill importance within an occupation, as annotated in the O*NET dataset, and our exposure measures. Using O*NET’s Basic Skills (skill definitions can be found in 5.2) and normalizing skill importance for each occupation, we attempted to enhance the interpretability of the results. We then perform regression analysis on our exposure measures ($E1$, $E1+0.5 \cdot E2$, $E1+E2$) to examine the associations between skill importance and exposure.

Our findings indicate a strong negative correlation between the importance of **science** and **critical thinking** skills and exposure, implying less likelihood of current LLMs impacting occupations requiring these skills. Conversely, **programming** and **writing** skills exhibit a strong positive association with exposure, indicating a higher exposure risk of occupations involving these skills to LLM influence (see Table 4 for detailed results).

5.2 O*NET Basic Skills Definitions

Basic Skills

Developed capacities that facilitate learning or the more rapid acquisition of knowledge.

Content

Background structures needed to work with and acquire more specific skills in a variety of different domains.

²Data is available here for the National Employment Matrix: <https://www.bls.gov/emp/tables/emp-by-detailed-occupation.htm>.

Basic Skill	E1 (std err)	E1 + 0.5 * E2 (std err)	E1 + E2 (std err)
<i>All skill importance scores are normalized to be between 0 and 1.</i>			
Constant	0.082*** (0.011)	-0.112*** (0.011)	0.300*** (0.057)
Active Listening	0.128** (0.047)	0.214*** (0.043)	0.449*** (0.027)
Mathematics	-0.127*** (0.026)	0.161*** (0.021)	0.787*** (0.049)
Reading Comprehension	0.153*** (0.041)	0.470*** (0.037)	-0.346*** (0.017)
Science	-0.114*** (0.014)	-0.230*** (0.012)	-0.346*** (0.017)
Speaking	-0.028 (0.039)	0.133*** (0.033)	0.294*** (0.042)
Writing	0.368*** (0.042)	0.467*** (0.037)	0.566*** (0.047)
Active Learning	-0.157*** (0.027)	-0.065** (0.024)	0.028 (0.032)
Critical Thinking	-0.264*** (0.036)	-0.196*** (0.033)	-0.129** (0.042)
Learning Strategies	-0.072* (0.028)	-0.209*** (0.025)	-0.346*** (0.034)
Monitoring	-0.067** (0.023)	-0.149*** (0.020)	-0.232*** (0.026)
Programming	0.637*** (0.030)	0.623*** (0.022)	0.609*** (0.024)

Table 4: Regression of occupation-level, human-annotated exposure to GPTs on skill importance for each skill in the O*NET Basic skills category, plus the programming skill.

- **Reading Comprehension** – Understanding written sentences and paragraphs in work-related documents.
- **Active Listening** – Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
- **Writing** – Communicating effectively in writing as appropriate for the needs of the audience.

658 • **Speaking** – Talking to others to convey information effectively.

659 • **Mathematics** – Using mathematics to solve problems.

660 • **Science** – Using scientific rules and methods to solve problems.

661 **Process**

662 Procedures that contribute to the more rapid acquisition of knowledge and skill across a variety
663 of domains

664 • **Critical Thinking** – Using logic and reasoning to identify the strengths and weaknesses of
665 alternative solutions, conclusions or approaches to problems.

666 • **Active Learning** – Understanding the implications of new information for both current
667 and future problem-solving and decision-making.

668 • **Learning Strategies** – Selecting and using training/instructional methods and procedures
669 appropriate for the situation when learning or teaching new things.

670 • **Monitoring** – Monitoring/Assessing performance of yourself, other individuals, or
671 organizations to make improvements or take corrective action.

672 **Cross-Functional Skills**

673 Note: We selected only Programming from the list of cross-functional skills because of our prior
674 knowledge about the models' ability to code.

675 • **Programming** – Writing computer programs for various purposes.

6 Education

6.1 Barriers to Entry

Next, we examine barriers to entry to better understand if there is differentiation in exposure due to types of jobs. One such proxy is an O*NET occupation-level descriptor called the "Job Zone." A Job Zone groups occupations that are similar in (a) the level of education needed to get a job in the occupation, (b) the amount of related experience required to do the work, and (c) the extent of on-the-job training needed to do the work. In the O*NET database, there are 5 Job Zones, with Job Zone 1 requiring the least amount of preparation (3 months) and Job Zone 5 requiring the most extensive amount of preparation, 4 or more years. We observe that median income increases monotonically across Job Zones as the level of preparation needed also increases, with the median worker in Job Zone 1 earning \$30,230 and the median worker in Job Zone 5 earning \$80,980.

All of our measures ($E1$, $E1 + 0.5 * E2$, and $E1 + E2$) show an identical pattern, that is, exposure increases from Job Zone 1 to Job Zone 4, and either remains similar or decreases at Job Zone 5. In Figure 8, we plot the percentage of workers at every threshold of exposure. We find that, on average, the percentage of workers in occupations with greater than 50 percent $E1 + 0.5 * E2$ exposure in Job Zones 1 through 5 have $E1 + 0.5 * E2$ at 0.00 percent (Job Zone 1), 6.11 percent (Job Zone 2), 10.57 percent (Job Zone 3), 34.5 percent (Job Zone 4), and 26.45 percent (Job Zone 5), respectively.

	Job Zone 1	Job Zone 2	Job Zone 3	Job Zone 4	Job Zone 5
Preparation	None or little (0-3 months)	Some (3-12 months)	Medium (1-2 years)	Considerable (2-4 years)	Extensive (4+ years)
Education	High school diploma or GED (optional)	High school diploma	Vocational school, on-the-job training, or associate's degree	Bachelor's degree	Master's degree or higher
Example Occupations	Food prep workers, dishwashers, floor sanders	Orderlies, customer service reps, tellers	Electricians, barbers, medical assistants	Database admins, graphic designers, cost estimators	Pharmacists, lawyers, astronomers
Median Income	\$30,230	\$38,215	\$54,815	\$77,345	\$81,980
Tot Emp (000s)	13,100	73,962	37,881	56,833	21,221
H E1	0.03	0.07	0.11	0.23	0.23
M E1	0.04	0.12	0.14	0.18	0.13
H E1 + 0.5 * E2	0.06	0.16	0.26	0.47	0.43
M E1 + 0.5 * E2	0.06	0.20	0.32	0.51	0.45
H E1 + E2	0.09	0.24	0.41	0.71	0.63
M E1 + E2	0.08	0.27	0.51	0.85	0.76

Table 5: Mean exposure to GPTs by job zone. For each job zone, we also present the median of median annual income for each constituting occupation in USD, and the total number of workers in all occupations for that job zone, in the thousands. "H" and "M" for each exposure type in column 1 refer to whether the exposure ratings were done by humans or the (GPT-4) model, respectively.

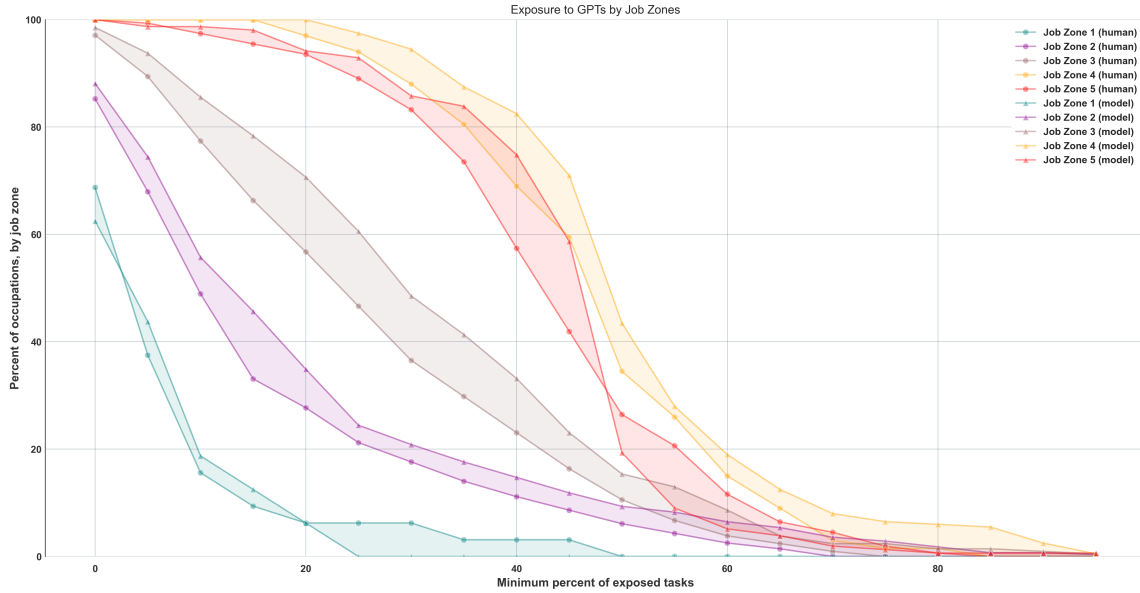


Figure 8: E1+0.5*E2 exposure ratings of occupations in the five Job Zones, which are groups of similar occupations that are classified according to the level of education, experience, and on-the-job training needed to perform them.

6.1.1 Typical Education Needed for Entry

Since inclusion in a Job Zone accounts for both the education required – which itself is a proxy for skill acquisition – and the preparation required, we seek data to disentangle these variables. We use two variables from the Bureau of Labor Statistics’ Occupational data: "Typical Education Needed for Entry" and "On-the-job Training Required to Attain Competency" in an occupation. By examining these factors, we aim to uncover trends with potential implications for the workforce. There are 3,504,000 workers for whom we lack data on education and on-the-job training requirements, and they are therefore excluded from the summary tables.

Our analysis suggests that individuals holding Bachelor’s, Master’s, and professional degrees are more exposed to LLMs and LLM-powered software than those without formal educational credentials (see Table 7). Interestingly, we also find that individuals with some college education but no degree exhibit a high level of exposure to LLMs and LLM-powered software. Upon examining the table displaying barriers to entry, we observe that the jobs with the least exposure require the most training. Conversely, jobs with no on-the-job training required or only internship/residency required appear to yield higher income but are more exposed to LLMs.

	Median Income	Emp (000s)	H $E1$	M $E1$	H $E1 + 0.5 * E2$	M $E1 + 0.5 * E2$	H $E1 + E2$	M $E1 + E2$
No formal educational credential	\$31,900	36,187	0.05	0.06	0.10	0.10	0.15	0.15
High school diploma or equivalent	\$45,470	67,033	0.09	0.13	0.20	0.25	0.31	0.37
Postsecondary nondegree award	\$48,315	9,636	0.07	0.15	0.19	0.28	0.31	0.41
Some college, no degree	\$40,970	2,898	0.23	0.34	0.39	0.53	0.55	0.72
Associate’s degree	\$60,360	3,537	0.12	0.14	0.31	0.36	0.49	0.59
Bachelor’s degree	\$78,375	71,698	0.23	0.17	0.47	0.51	0.70	0.84
Master’s degree	\$79,605	3,216	0.26	0.14	0.46	0.44	0.66	0.74
Doctoral or professional degree	\$82,420	5,290	0.21	0.13	0.41	0.43	0.60	0.74

Table 6: Mean exposure scores for occupations, grouped by typical education needed for entry into the occupation. Alongside exposure scores, we display the median of median annual income for each occupation, as well as the total number of workers in each group, in thousands. "H" and "M" for each exposure type in the top row refers to whether the exposure ratings were done by humans or the (GPT-4) model, respectively.

On The Job Training Required	None	Apprenticeship	Internship or Residency	Short-term	Moderate-term	Long-term
Median Income	\$77,440	\$55,995	\$77,110	\$33,370	\$46,880	\$48,925
Tot Emp (000s)	90,776	3,066	3,063	66,234	31,285	5,070
H E1	0.20	0.01	0.16	0.11	0.09	0.08
M E1	0.16	0.02	0.06	0.15	0.12	0.10
H E1 + 0.5 * E2	0.42	0.04	0.36	0.21	0.21	0.18
M E1 + 0.5 * E2	0.46	0.06	0.38	0.25	0.25	0.22
H E1 + E2	0.63	0.07	0.55	0.32	0.32	0.28
M E1 + E2	0.76	0.10	0.71	0.34	0.38	0.33

Table 7: Mean exposure scores for occupations, grouped by level of on-the-job training required to attain competency in the job. Alongside exposure scores, we display the median of median annual income for each occupation, as well as the total number of workers in each group, in thousands. "H" and "M" for each exposure type in column 1 refer to whether the exposure ratings were done by humans or the (GPT-4) model, respectively.

7 Industrial and Productivity Exposure

Figures 9 and 10 show the overall employment-weighted relative exposure of 3-digit NAICS industries according to human raters and GPT-4 respectively (based on our exposure rubric). The impact potential is present across nearly all industries, with wide heterogeneity. Both methods agree generally on relative exposures: data processing, information processing, and hospitals all have high exposure.

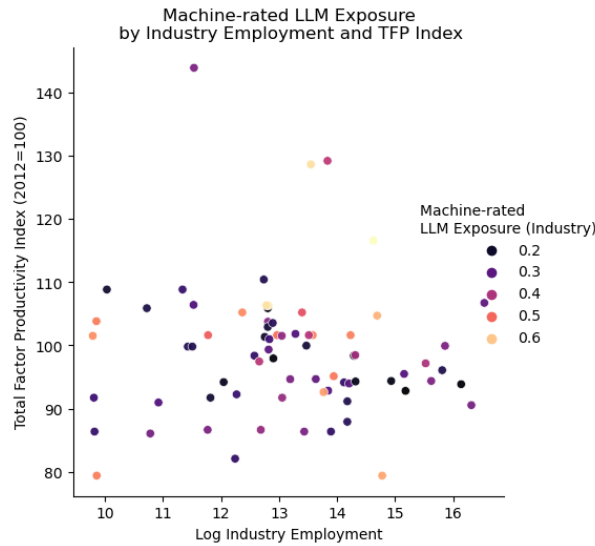


Figure 11: Machine-rated LLM Exposure Industry Total Factor Productivity Scatter

Figure Notes: Total Factor Productivity level as of 2020 is on the y-axis and logged industry employment is on the x-axis. Points are industries (3-digit NAICS) colored by their GPT4-rated exposure, weighing occupational exposures by total employed in each 6-digit SOC Code. There is little correlation between change in productivity level since 2012 and exposure.

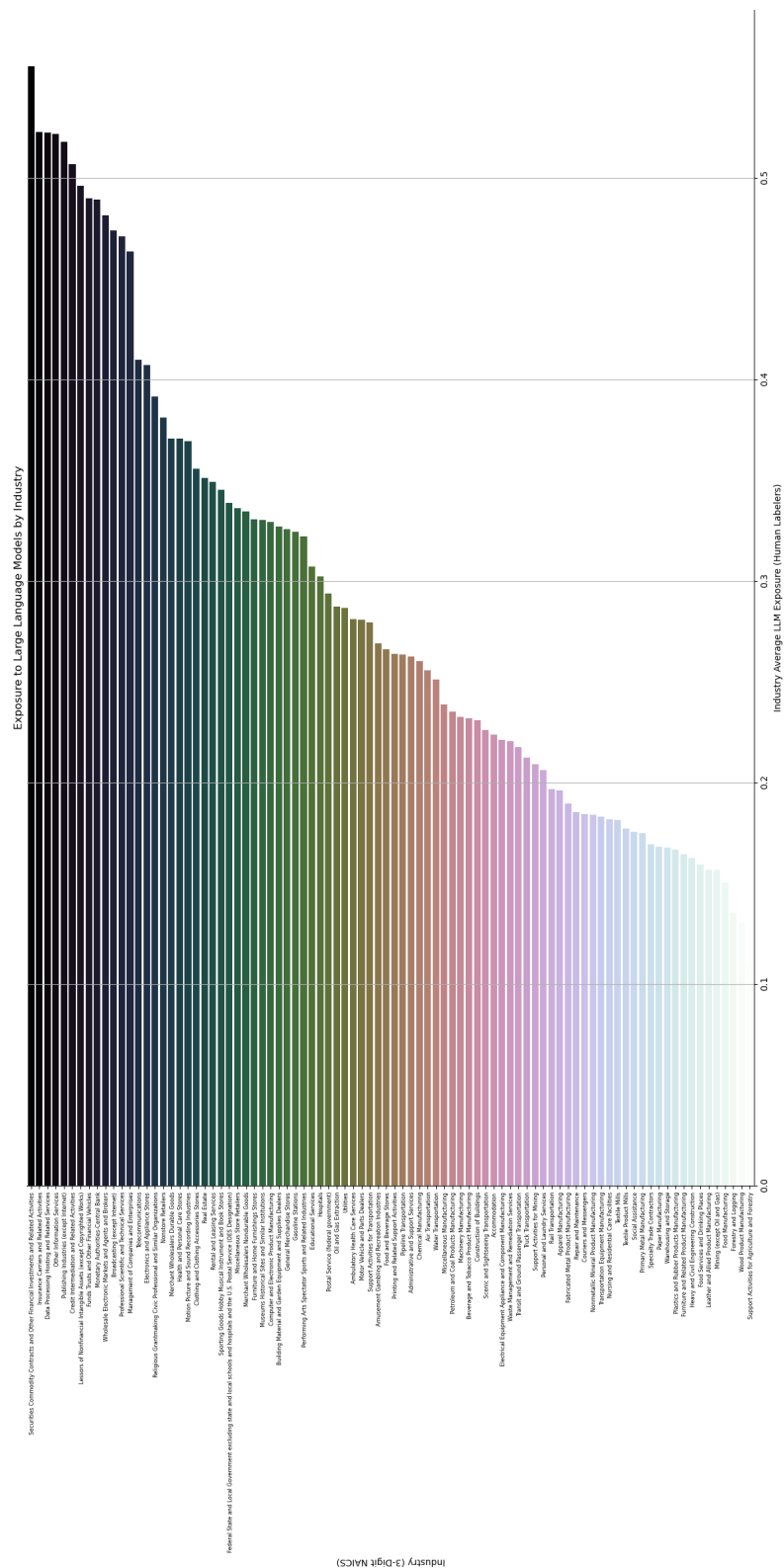


Figure 9: Human-labeled exposure aggregated up to the 3-digit NAICS industry. Exposure-levels are derived from the E1+0.5*E2 score set at the task-level, then aggregated to occupations. Industries measures are built from employment-weights at the SOC6-level within 3-digit NAICS.

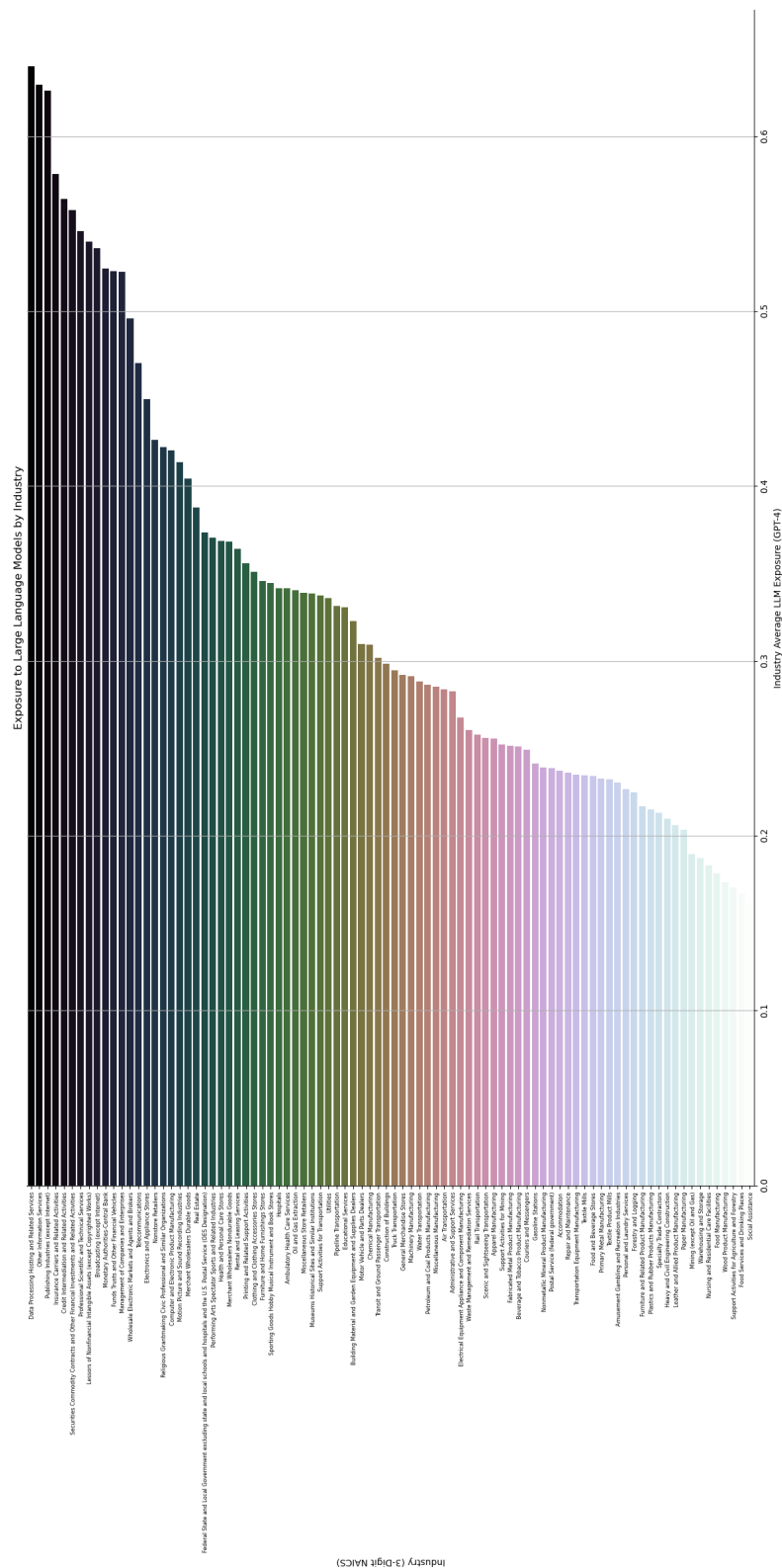


Figure 10: GPT4-labeled exposure aggregated up to the 3-digit NAICS industry. Exposure-levels are derived from the E1+0.5*E2 scoreset at the task-level, then aggregated to occupations. Industries measures are built from employment-weights at the SOC6-level within 3-digit NAICS.

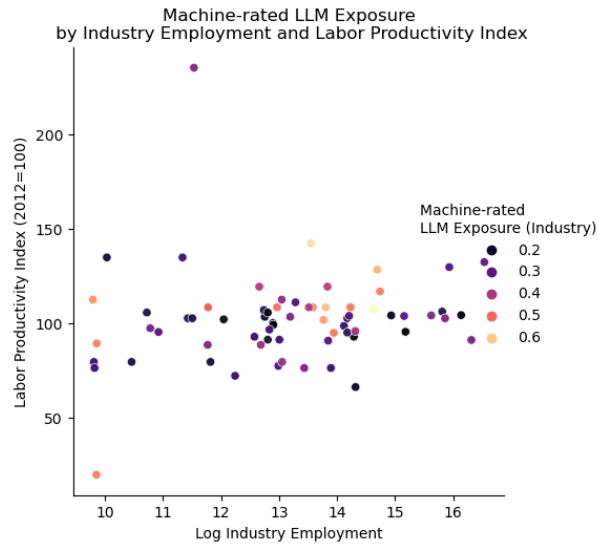


Figure 12: Machine-rated LLM Exposure Industry Labor Productivity Scatter

Labor Productivity level as of 2020 is on the y-axis and logged industry employment is on the x-axis. Points are industries (3-digit NAICS) colored by their GPT4-rated exposure, weighing occupational exposures by total employed in each 6-digit SOC Code. There is little correlation between change in productivity level since 2012 and exposure.

Recent productivity growth (both total factor and labor) appears uncorrelated with exposure as well. Figures 11 and 12 show little relationship between productivity growth as of 2020 since 2012 and current exposure to LLMs as rated by the model. Productivity growth since 2012 and exposure to LLM technologies appear unrelated. Productivity data are available from <https://www.bea.gov/data/special-topics/integrated-industry-level-production-account-klems>.

8 Exposed Occupations

8.1 Occupations Without Any Exposed Tasks

Occupations with no labeled exposed tasks

Agricultural Equipment Operators
 Athletes and Sports Competitors
 Automotive Glass Installers and Repairers
 Bus and Truck Mechanics and Diesel Engine Specialists
 Cement Masons and Concrete Finishers
 Cooks, Short Order
 Cutters and Trimmers, Hand
 Derrick Operators, Oil and Gas
 Dining Room and Cafeteria Attendants and Bartender Helpers
 Dishwashers
 Dredge Operators
 Electrical Power-Line Installers and Repairers
 Excavating and Loading Machine and Dragline Operators, Surface Mining
 Floor Layers, Except Carpet, Wood, and Hard Tiles
 Foundry Mold and Coremakers
 Helpers—Brickmasons, Blockmasons, Stonemasons, and Tile and Marble Setters
 Helpers—Carpenters
 Helpers—Painters, Paperhangers, Plasterers, and Stucco Masons
 Helpers—Pipelayers, Plumbers, Pipefitters, and Steamfitters
 Helpers—Roofers
 Meat, Poultry, and Fish Cutters and Trimmers
 Motorcycle Mechanics
 Paving, Surfacing, and Tamping Equipment Operators
 Pile Driver Operators
 Pourers and Casters, Metal
 Rail-Track Laying and Maintenance Equipment Operators
 Refractory Materials Repairers, Except Brickmasons
 Roof Bolters, Mining
 Roustabouts, Oil and Gas
 Slaughterers and Meat Packers
 Stonemasons
 Tapers
 Tire Repairers and Changers
 Wellhead Pumpers

Table 8: All 34 occupations for which none of our measures labeled any tasks as exposed.

8.2 Occupations With Greatest Exposure

Group	Occupations with highest exposure	% Exposure
Human $E1$	Interpreters and Translators	76.5
	Survey Researchers	75.0
	Poets, Lyricists and Creative Writers	68.8
	Animal Scientists	66.7
	Public Relations Specialists	66.7
Human $E1 + 0.5 * E2$	Survey Researchers	84.4
	Writers and Authors	82.5
	Interpreters and Translators	82.4
	Public Relations Specialists	80.6
	Animal Scientists	77.8
Human $E1 + E2$	Mathematicians	100.0
	Tax Preparers	100.0
	Financial Quantitative Analysts	100.0
	Writers and Authors	100.0
	Web and Digital Interface Designers	100.0
	<i>Humans labeled 15 occupations as "fully exposed."</i>	
Model $E1$	Mathematicians	100.0
	Correspondence Clerks	95.2
	Blockchain Engineers	94.1
	Court Reporters and Simultaneous Captioners	92.9
	Proofreaders and Copy Markers	90.9
Model $E1 + 0.5 * E2$	Mathematicians	100.0
	Blockchain Engineers	97.1
	Court Reporters and Simultaneous Captioners	96.4
	Proofreaders and Copy Markers	95.5
	Correspondence Clerks	95.2
Model $E1 + E2$	Accountants and Auditors	100.0
	News Analysts, Reporters, and Journalists	100.0
	Legal Secretaries and Administrative Assistants	100.0
	Clinical Data Managers	100.0
	Climate Change Policy Analysts	100.0
	<i>The model labeled 86 occupations as "fully exposed."</i>	
Highest variance	Search Marketing Strategists	14.5
	Graphic Designers	13.4
	Investment Fund Managers	13.0
	Financial Managers	13.0
	Insurance Appraisers, Auto Damage	12.6

Table 9: Occupations with the highest exposure according to each measurement

The bottom section lists the occupations with the highest σ^2 value, indicating that they had the most variability in exposure scores. Exposure percentages indicate the share of an occupation's task that are exposed to GPTs ($E1$) or GPT-powered software ($E1 + 0.5 * E2$ and $E1 + E2$), where exposure is defined as driving a reduction in time it takes to complete the task by at least 50%. As such, occupations listed in this table are those where we estimate that GPTs and GPT-powered software are able to save workers a significant amount of time completing a large share of their tasks, but it does not necessarily suggest that their tasks can be fully automated by these technologies.

We create a network of occupations by taking each of the O*NET-SOC occupations and representing them as a node connected to other occupations if they share a detailed work activity (DWA) in common. We then cluster the network using the Louvain method and compute the network modularity (0.709), revealing 11 communities. The Louvain method is an efficient and popular algorithm used for community detection in large networks. It detects communities by maximizing modularity, in this case, the density of shared activities inside of clusters of jobs relative to activities shared outside of clusters of jobs. A perfectly modular network will have a modularity measure of 1, while a perfectly non-modular network has a measure of -0.5. The network is below in Figure 13 with node communities having the same color. A labeled version of the same plot is also below in Figure 14. Pictured below as well is the network of detailed work activities (nodes) connected by occupations that they share. This network is also highly modular (0.768 Louvain modularity).



Figure 13: Network Communities of Occupations

Figure Notes: Occupation nodes are connected by an edge if they share a detailed work activity (DWA). The Louvain Method is used to detect communities and the occupation network has a modularity score of 0.709.

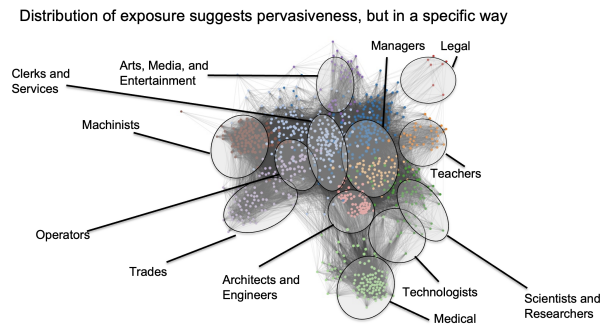


Figure 14: Network Communities of Occupations with Labels

Figure Notes: Occupation nodes are connected by an edge if they share a detailed work activity (DWA). The Louvain Method is used to detect communities and the occupation network has a modularity score of 0.709.

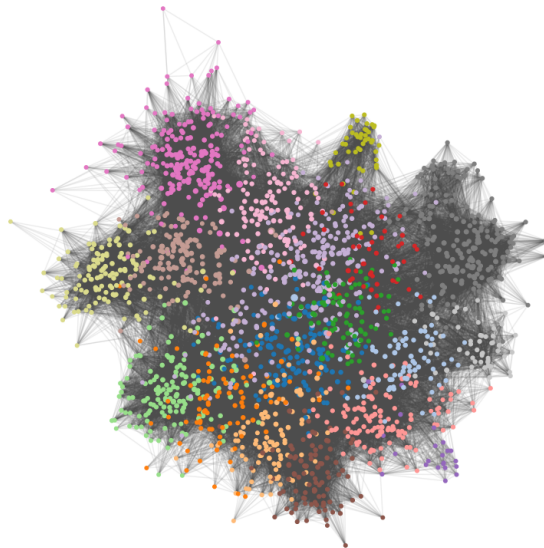


Figure 15: Network Communities of Detailed Work Activities

Figure Notes: This is the inverse of the occupation network plot. In this network, nodes are detailed work activities (DWAs), connected if there's an occupation that shares the two activities. The Louvain Method is used to detect communities and the DWA network has a modularity score of 0.768. This is slightly higher than the occupational network modularity. There are 18 detailed work activity clusters. Clusters of DWAs are generated in the supplementary materials files.

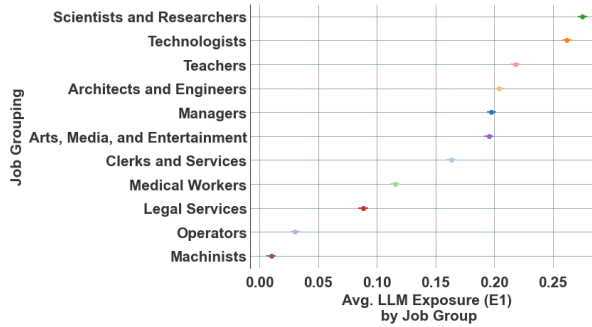


Figure 16: Exposure by Occupation Group

Figure Notes: E1 Exposure by job group. Scientists and Researchers, as well as Technologists, have the highest share of their tasks exposed to LLMs of all job groups (according to our human-based E1 measure).

9 Validation of Measures

9.1 Comparison to Earlier Efforts

We compare our measurements to previous efforts documenting the distribution of automation exposure in the economy and find broadly consistent results. Most other technology exposure measures we examine are statistically significantly correlated with our preferred exposure measure, while measures of manual routineness and robotics exposure show negative correlations. The variance explained by these earlier efforts (34, 36, 39, 42, 43, 78), along with wage controls, ranges from 60 to 72 percent, indicating that 28 to 40 percent of the variation in our AI exposure measure remains unaccounted for by previous technology exposure measurements.

This paper aims to build on a number of previous empirical studies examining the occupational exposure to advances in AI and/or automation. Previous studies have used a variety of methods, including:

- Using occupational taxonomies like O*NET to characterize which occupations have routine vs. non-routine and manual vs. cognitive task content (24, 78).
- Mapping text descriptions of tasks to descriptions of technological advances in patents.

(34, 79)

- Linking capabilities of AI systems to occupational abilities and aggregating exposure estimates to the occupations where those abilities are required. (36, 37)
- Mapping the results of AI task benchmark evaluations (ImageNet, Robocup, etc.) to 59 worker tasks through a set of 14 cognitive abilities drawn from the cognitive science literature. (38)
- Expert labeling of automation potential for a set of O*NET occupations where experts had high confidence, combined with a probabilistic classifier to estimate automation potential for the remainder of O*NET occupations. (39)
- Developing a rubric for evaluating the "suitability for machine learning" (SML) of activities that workers are completing in the economy (42, 43, 73).

We provide a set of summary statistics on many of these prior efforts in Table 10.

This paper's methodology primarily builds upon the SML approach by developing a rubric to evaluate the overlap between LLM capabilities and worker tasks as reported in the O*NET database. We find that earlier measures from (25, 34, 36, 39, 42, 43) combined are predictive of LLM exposure. Table 11 presents the results of OLS regressions of our new LLM exposure measurements on occupation-level exposure measures from (36) ("AI Occupational Exposure Score" in the table), (39) (Frey & Osborne Automation), scores from all three technologies in (34), normalized routine manual and cognitive scores from (78), and (42, 43) (SML). We also use annualized occupational salaries from the most recent BLS Occupational Employment Survey as a control. There are four separate output variables representing new scores in this paper that are predicted by earlier efforts.

GPT-4 Exposure Rating 1 corresponds to our overall exposure rubric as evaluated by GPT-4,

774 where full exposure potential is coded as 1, no exposure potential is coded as 0, and partial
775 exposure (E2 in our labeling scheme) is coded as 0.5. GPT-4 Exposure Rating 2 is scored similarly
776 for overall exposure, but with a slightly different prompt. The results are very similar across the
777 two prompts. Human Exposure Rating represents the same rubric as in GPT-4 Exposure Rating 1
778 but is scored by humans, as discussed in an earlier section of the paper. These results correspond
779 to the $E1 + 0.5 * E2$ set of statistics presented above.

780 The results across each type of measurement are consistent. We find generally positive
781 and statistically significant correlations between our LLM exposure measures and previous
782 measurements targeting software and AI. Encouragingly, the SML exposure scores by occupation
783 show significant and positive associations with the exposure scores we develop in this paper,
784 demonstrating a level of cohesion between the two studies with similar approaches. The Webb
785 software and AI patent-based measures, SML, and normalized (demeaned and divided by standard
786 deviation) routine cognitive scores all exhibit positive associations with some of our measures.
787 AI Occupational Exposure from (36) is the most correlated in a bivariate regression with our
788 scores but not statistically significant in the multivariate case. Overall these prior efforts explain
789 60 to 70 percent of the variation in LLM exposure.

790 Software, SML, and routine cognitive scores all show positive and statistically significant
791 associations with LLM exposure scores at a 1 percent level. Coefficients on AI scores from (34)
792 are also positive and statistically significant at a 5 percent level, but our secondary prompt
793 on overall exposure to LLMs in columns 3 and 4 does not exhibit a statistically significant
794 relationship. For the most part, the AI Occupational Exposure Score is not correlated with
795 our exposure measures. Webb's Robot exposure scores, routine manual task content, and the
796 overall Automation metric from (39) are all negatively correlated with our primary GPT-4 and
797 human-assessed overall exposure ratings, conditional on the other measurements. This negative
798 correlation reflects the limited exposure of physical tasks to LLMs. Manual work is not exposed

799 to LLMs or even LLMs with additional systems integration for the time being.

800 Low correlations with (36) and (39) could potentially be explained by differences in approaches.
801 Linking AI capabilities to worker abilities or scoring exposure directly based on the occupation's
802 characteristics, rather than aggregating up to the occupation from DWA or task-level scoring (as in
803 the SML paper and our own), offer a slightly different perspective on the content of occupations.

804 In all regressions, the R^2 ranges between 60.7 percent (column 3) and 72.8 percent (column
805 5). This suggests that our measure, which explicitly focuses on LLM capabilities, has between 28
806 and 40 percent unexplained variance compared to other measurements. Particularly in the case
807 of AI-related exposure scores, we anticipate that a combination of other measurements would
808 have a strong correlation with our scores. However, earlier efforts had limited information about
809 the future progress of LLMs or LLM-powered software. We expect that our understanding of
810 future machine learning technologies is similarly imperfectly captured by our rubric today.

811 Individually, scores related to routine manual or robotics exposures are negatively correlated
812 with LLM exposure ($E1 + 0.5 * E2$), while AI-related exposure scores are positively correlated
813 with LLM exposure. A one standard deviation increase in the AI score from (34) or the Suitability
814 for Machine Learning score described in (42, 43) is associated with a respective 0.024 and 0.055
815 standard deviation increase in LLM exposure from the GPT-4 Ratings conditional on other scores.

	Min	25th Perc.	Median	75th Perc	Max	Mean	Std. Dev.	Count
GPT-4 Exposure Rating 1	0.00	0.13	0.34	0.50	1.00	0.33	0.22	750
GPT-4 Exposure Rating 2	0.00	0.09	0.24	0.40	0.98	0.26	0.20	750
Human Exposure Rating	0.00	0.09	0.29	0.47	0.84	0.29	0.21	750
Software (Webb)	1.00	25.00	50.00	75.00	100.00	50.69	30.05	750
Robot (Webb)	1.00	22.00	52.00	69.00	100.00	48.61	28.61	750
AI (Webb)	1.00	28.00	55.00	82.00	100.00	54.53	29.65	750
Suitability for Machine Learning	2.60	2.84	2.95	3.12	3.55	2.99	0.18	750
Normalized Routine Cognitive	-3.05	-0.46	0.10	0.63	3.42	0.07	0.86	750
Normalized Routine Manual	-1.81	-0.81	-0.11	0.73	2.96	0.05	1.01	750
AI Occupational Exposure Score	1.42	3.09	3.56	4.04	6.54	3.56	0.70	750
Frey & Osborne Automation	0.00	0.07	0.59	0.88	0.99	0.50	0.38	681
Log Avg. Salary	10.13	10.67	11.00	11.34	12.65	11.02	0.45	749

Table 10: Summary statistics for a suite of prior efforts to measure occupational exposure to AI and automation. We have also included summary statistics for measurements newly presented in this work. We include all measures from (34), normalized routine cognitive and manual scores from (78) (means may deviate slightly from 0 due to imperfect matching of occupational groups), Suitability for Machine Learning from (42, 43, 73), AI Occupational Exposure from (36), and Automation exposure from (39). We include as many occupations as we can match, but since O*NET taxonomies have changed as these measures have been developed, some of the roles may be missing from the most recent version of O*NET 6-digit occupations.

	GPT-4 Exposure Rating 1		GPT-4 Exposure Rating 2		Human Exposure Rating	
	(1)	(2)	(3)	(4)	(5)	(6)
Software (Webb)	0.03406*** (0.00942)	0.03705*** (0.00942)	0.03347*** (0.00940)	0.03580*** (0.00935)	0.02893*** (0.00940)	0.03032*** (0.00945)
Robot (Webb)	-0.10799*** (0.00909)	-0.11568*** (0.00877)	-0.10777*** (0.00982)	-0.11399*** (0.00941)	-0.10609*** (0.00834)	-0.10957*** (0.00806)
AI (Webb)	0.02370*** (0.00897)	0.02667*** (0.00871)	0.01062 (0.00896)	0.01330 (0.00875)	0.01996** (0.00902)	0.02116** (0.00878)
Suitability for Machine Learning	0.05445*** (0.00831)	0.04959*** (0.00815)	0.05250*** (0.00812)	0.04841*** (0.00801)	0.03599*** (0.00736)	0.03389*** (0.00717)
Routine Cognitive	0.05663*** (0.00760)	0.05892*** (0.00767)	0.04069*** (0.00749)	0.04303*** (0.00754)	0.03061*** (0.00576)	0.03139*** (0.00574)
Routine Manual	-0.11213*** (0.00790)	-0.11439*** (0.00794)	-0.09446*** (0.00821)	-0.09618*** (0.00823)	-0.11111*** (0.00745)	-0.11218*** (0.00748)
AI Occupational Exposure Score	0.00695 (0.00775)	0.01726** (0.00741)	-0.01077 (0.00812)	-0.00186 (0.00780)	0.00441 (0.00643)	0.00877 (0.00592)
Frey & Osborne Automation	-0.01143* (0.00694)	-0.01493** (0.00696)	-0.00138 (0.00759)	-0.00460 (0.00746)	-0.01471** (0.00712)	-0.01608** (0.00702)
Log Avg. Salary	0.05804*** (0.01870)		0.04863*** (0.01860)		0.02531 (0.01727)	
Constant	-0.31072 (0.20590)	0.32778*** (0.00493)	-0.27579 (0.20502)	0.25903*** (0.00520)	0.01717 (0.19008)	0.29577*** (0.00468)
N	680	681	680	681	680	681
R ²	0.68741	0.68212	0.60737	0.60198	0.71213	0.71126

Table 11: Regression of LLM-exposure scores on prior measures of occupational exposure to AI and automation. We also include annualized wages from the BLS-OES survey in May 2021. Each measure is standardized to mean zero and variance 1. Generally we find strong positive associations with previous efforts, though large residual variance to still be explained by our new measures. Columns 1 and 2 are based on our main $E1+0.5 \cdot E2$ exposure measure from GPT-4 ratings. Columns 3 and 4 are based on a similar slightly different exposure rubric also rated by GPT-4 for robustness. Columns 5 and 6 reflect human ratings on the same rubric as columns 1 and 2.

	GPT-4 Exposure Rating 1		GPT-4 Exposure Rating 2		Human Exposure Rating	
	(1)	(2)	(3)	(4)	(5)	(6)
Software (Webb)	0.00113*** (0.00031)	0.00123*** (0.00031)	0.00111*** (0.00031)	0.00119*** (0.00031)	0.00096*** (0.00031)	0.00101*** (0.00031)
Robot (Webb)	-0.00378*** (0.00032)	-0.00405*** (0.00031)	-0.00377*** (0.00034)	-0.00399*** (0.00033)	-0.00371*** (0.00029)	-0.00383*** (0.00028)
AI (Webb)	0.00080*** (0.00030)	0.00090*** (0.00029)	0.00036 (0.00030)	0.00045 (0.00030)	0.00067** (0.00030)	0.00071** (0.00030)
Suitability for Machine Learning	0.29522*** (0.04503)	0.26888*** (0.04418)	0.28468*** (0.04404)	0.26245*** (0.04342)	0.19514*** (0.03990)	0.18373*** (0.03886)
Normalized Routine Cognitive	0.06601*** (0.00886)	0.06868*** (0.00894)	0.04743*** (0.00872)	0.05015*** (0.00879)	0.03568*** (0.00671)	0.03659*** (0.00669)
Normalized Routine Manual	-0.11147*** (0.00785)	-0.11371*** (0.00789)	-0.09390*** (0.00817)	-0.09561*** (0.00818)	-0.11045*** (0.00741)	-0.11152*** (0.00744)
AI Occupational Exposure Score	0.00993 (0.01107)	0.02465** (0.01059)	-0.01537 (0.01160)	-0.00265 (0.01114)	0.00630 (0.00918)	0.01252 (0.00845)
Frey & Osborne Automation	-0.03024* (0.01835)	-0.03950** (0.01841)	-0.00364 (0.02007)	-0.01217 (0.01972)	-0.03890** (0.01883)	-0.04253** (0.01858)
Log Avg. Salary	0.05804*** (0.01870)		0.04863*** (0.01860)		0.02531 (0.01727)	
Constant	-1.12937*** (0.26859)	-0.45743*** (0.15327)	-0.96117*** (0.26365)	-0.39935*** (0.15017)	-0.47078* (0.24684)	-0.17706 (0.13256)
N	680	681	680	681	680	681
R ²	0.68741	0.68212	0.60737	0.60198	0.71213	0.71126

Table 12: Regression of LLM-exposure scores on prior measures of occupational exposure to AI and automation. We also include annualized wages from the BLS-OES survey in May 2021. Each measure is kept in its original scale, with the exception of routine cognitive and routine manual scores from (78). Those two scores are standardized to mean zero and variance 1. Generally we find strong positive associations with previous efforts, though large residual variance to still be explained by our new measures. Columns 1 and 2 are based on our main $E1+0.5 \cdot E2$ exposure measure from GPT-4 ratings. Columns 3 and 4 are based on a similar slightly different exposure rubric also rated by GPT-4 for robustness. Columns 5 and 6 reflect human ratings on the same rubric as columns 1 and 2.

9.2 Survey Results with ChatGPT Users

Validating our rubric for measuring LLM task exposure is difficult due to the technology's emerging nature. However, we attempted to validate our exposure results by comparing them to reported real-world LLM usage by surveying users on the ChatGPT Discord in February 2023. We asked them to identify their work-related uses, and matched those use cases to the most closely related O*NET tasks (see Table 13). Respondents opted in to participating in the survey, are from a range of countries around the globe. The survey asked members whether they used ChatGPT in their current jobs and specifically asked them, "What are some ways that you use ChatGPT in your work? Please list and describe up to three." Using the OpenAI API, we retrieved the embeddings for the tasks that survey respondents report using ChatGPT for in their work. We then compared the cosine similarity of those embeddings matrices to the embeddings for all O*NET tasks, identifying the single O*NET task that is most similar to each user-reported task.

The table below presents the most-reported O*NET tasks that respondents reported using ChatGPT for, as well as the GPT-4 and human annotations made using our exposure rubric. Human annotations marked two of the top 30 tasks as E0 (i.e. no exposure) and the GPT-4 labels marked one task as such. Overall, humans scored 13 out of the top 30 tasks as E1 (exposed to an LLM like ChatGPT directly) and 15 of the tasks as E2 (exposed to an LLM + additional software). GPT-4 was much more likely to rate these top tasks as E1 than E2, rating 24 of the 30 tasks as E1 and only 5 as E2. Most tasks identified by Discord respondents aligned with our rubric's exposure levels: 5 percent were rated (E0), 60 percent exposed (E1), and 35 percent exposed with additional software integration (E2). A t-SNE visualization (Figure 17) confirms this, showing few real-world LLM tasks in low-exposure areas. Although the data, based on brief text snippets, was noisy, the survey results support our rubric-based findings.

Table 13: Job Categories for Discord Survey Respondents

Job Category	Count
Software Eng/Dev/Game Developer/Programmer	259
Other	225
Manager/Executive/Entrepreneur/Business Owner	120
Student	84
Educator	69
Writer	56
Marketing/Advertising	40
Designer	29
Researcher	27
Visual Arts	21
Data Scientist or Analyst	20
Healthcare	17
Sales	13
Finance	9
Customer Service	4
Real Estate	3
Total Responses	877

Task	GPT-4 Label	Human Label	count
Research new software or design concepts.	E2	E2	45
Compose and edit scripts or provide screenwriters with story outlines from which scripts can be written.	E1	E1	20
Help children with homework and school work.	E1	E1	20
Compose explanatory summaries of contents of books and other reference materials.	E1	E1	17
Write and execute scripts to automate tasks, such as parsing large data files.	E1	E2	14
Answer students' questions.	E1	E0	14
Develop or implement data analysis algorithms.	E1	E2	13
Answer user inquiries regarding computer software or hardware operation to resolve problems.	E2	E1	12
Train robots, using artificial intelligence software or interactive training techniques, to perform simple or complex tasks, such as designing and carrying out a series of iterative tests of chemical samples.	E1	E2	12
Write online blog entries that address news developments or offer additional information, opinions, or commentary on news events.	E1	E1	12
Write algorithms or programming code for ad hoc robotic applications.	E1	E2	11
Write new functions or applications in programming languages to conduct analyses.	E1	E1	11
Write computer programs or scripts to be used in querying databases.	E1	E1	11
Compile and write documentation of program development and subsequent revisions, inserting comments in the coded instructions so others can understand the program.	E1	E2	10
Maintain understanding of current Web technologies or programming practices through continuing education, reading, or participation in professional conferences, workshops, or groups.	E2	E2	10
Write supporting code for Web applications or Web sites.	E1	E2	9
Conduct market research analysis to identify search query trends, real-time search and news media activity, popular social media topics, electronic commerce trends, market opportunities, or competitor performance.	E2	E2	8
Write articles and books.	E1	E1	8
Prepare, rewrite and edit copy to improve readability, or supervise others who do this work.	E1	E1	8
Optimize Web site exposure by analyzing search engine patterns to direct online placement of keywords or other content.	E2	E2	8
Post and update content on the company's Web site and social media outlets.	E1	E1	7
Write articles, bulletins, sales letters, speeches, and other related informative, marketing and promotional material.	E1	E1	7
Read copy or proof to detect and correct errors in spelling, punctuation, and syntax.	E1	E1	7
Conduct searches to find needed information, using such sources as the Internet.	E1	E2	6
Design or develop automated testing tools.	E1	E1	6
Design and create special programs for tasks such as statistical analysis and data entry and cleaning.	E1	E2	5
Present drafts and ideas to clients.	E1	E2	5
Discuss the design with the client.	E0	E2	5
Compose, type, and distribute meeting notes, routine correspondence, or reports, such as presentations or expense, statistical, or monthly reports.	E1	E0	5
Collaborate with computer manufacturers and other users to develop new programming methods.	E1	E2	5

Table 14: Top 30 most reported O*NET tasks that OpenAI discord survey respondents report using ChatGPT for in their work. There were a total of 954 survey responses.

The task mappings are of course relatively brittle – the embeddings are based on short text strings without much context. This can create some noise in both the visualization and the ratings similarity of tasks. Such noise should bias our results toward finding less overlap between our task-level scores and the scores for the survey-linked tasks.

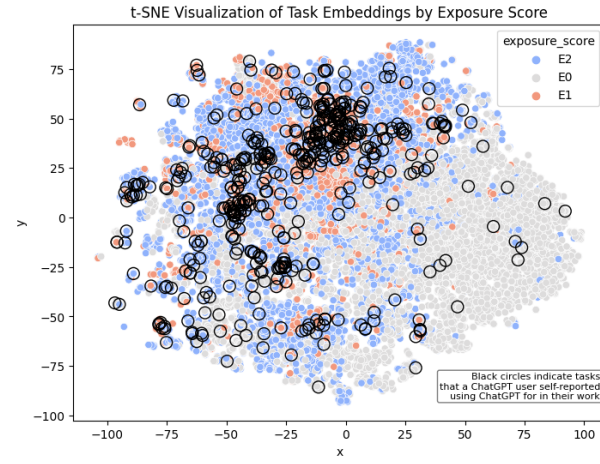


Figure 17

Embeddings for all O*NET tasks plotted in 2-D space and colored according to the GPT-4 exposure score annotations for the given task. Black circles denote the 460 unique O*NET tasks whose text embeddings were most similar to the tasks that a sample of 954 ChatGPT users self-reported using ChatGPT for in their work. As you can see, these circles primarily overlap with tasks labeled E1 and E2.

9.3 Alternative Weightings

Most of our analysis uses one of two task-weighting approaches. The default approach is "core-weighting", where we assign twice the weight to a task within an occupation if the task is labeled as "core" than if it is labeled as "supplemental" or not labeled in the O*NET task database. That is, each core task receives a weight of 2 and each supplemental or unlabeled task receives as weight of 1. We then normalize those weights within occupation by the occupation-weight sum such that all weights now sum to one. The other approach used in some analyses (and indicated as such) is equal-weighting, where all tasks receive the same weight within an occupation.

We then aggregate scores from tasks to occupations or from occupations to industries by taking a weighted average of the rubric scores. The precise assignment of scores (e.g. E1 is mapped to 1, E0 is mapped to 0) is indicated for the analysis component in question. We also consider using the importance scores in the O*NET database (labeled as "IM"). Following a similar procedure to the core-weighting, we use the importance weights and normalize them

by their within-occupation sum, creating a task-weight to apply to our rubric measures. There are some tasks in O*NET that are missing importance scores, reducing the sample of available occupations from 923 for core or equal weighting to 873 with importance weighting. Nevertheless, the correlation between occupation-level scores for both GPT-4 and Human rubric ratings is 0.997. We find that using importance scores instead of core or equal weights does little to change occupation-level results. As a result we omit the importance weights from the main analysis. Figure 18 below shows the scatterplot of occupation scores for the E1+0.5E2 score values. Code available in the Github repository facilitates using importance scores instead if desired.

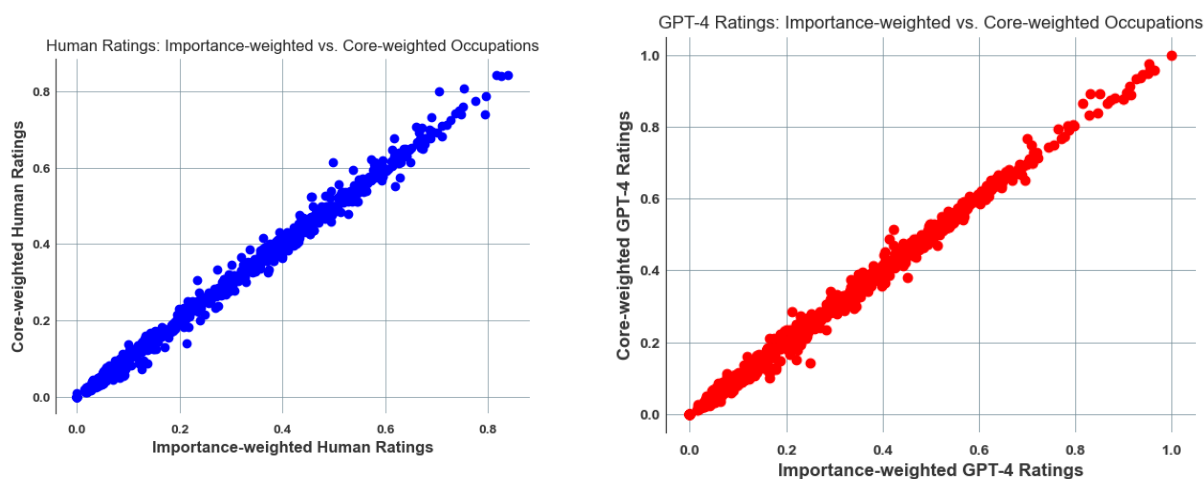


Figure 18: Scatterplot of Occupation-level Ratings. Here we examine occupation-level score differences using two different weighting schemes for tasks. In core-weighted tasks, apply a weight of 2 for all tasks designated as "core" in the O*NET database, and otherwise apply a weight of 1 (for supplemental or otherwise not classified tasks). We normalize these weights within occupation so that they sum to one. In importance-weighted tasks, we use the "IM" score in the O*NET database for each task as its weight, again normalizing the weights within occupation so that they sum to one. These weights at the task-level are then multiplied by the exposure score metric for either human or GPT-4 ratings. E0 is treated as a 0, E1 is treated as a 1.0, and E2 is a 0.5. Multiplying the weights on tasks within occupations and summing over all tasks, we get a weighted-average occupation-level score. These scores are plotted against each other in the scatterplot. The two measures under different weighting schemes have a correlation of 0.997 for both human and GPT-4 comparisons. However, missing data means that importance scores are only available for 873 occupations, while 923 have available core or supplemental categories. We either choose core-weighting or equal weighting of tasks in our main analyses.

10 Early evidence of LLMs Spawning Complementary Software Development

10.1 Analysis of LLM Impact on Developer Activity Using Github Forks Data

Understanding the impact of LLM deployment on software development is critical to assessing their potential as GPTs. These additional innovations are already extending the reach of LLM capabilities, including plug-ins for GPT-4 (80), Retrieval-Augmented Generation tools to improve the reference materials for LLM queries (17), and fine-tuning that increases the quality of contextual responses (16). To test the hypothesis that the release of ChatGPT caused an increase in development activity, we obtained data on forks over time from 1,167 public Github repositories that, based on the content of their Readme files, descriptions, and code, can be reasonable assumed to be differentially impacted by the release of a LLM like ChatGPT. We then use a synthetic difference-in-difference approach (as in (81)) to estimate the causal impact of ChatGPT release on weekly Github forks in impacted repositories. Forks are a feature of the Github platform that allows users to create a personal copy of someone else's repository. In practice, this is often useful when a user wants to use the existing code as a starting point for their own work without directly modifying the original repository. We hypothesize that repositories focused on LLM-related coding projects should see a large spike in forks after the release of ChatGPT relative to those developing code for statistical analysis and data science purposes.

In order to identify a set of repositories that we hypothesize would be heavily impacted by the release of ChatGPT, we first attempted to scrape forks data from the 600 most forked repositories with the query terms "LLM" or "language model" or "code generation" or "LaMDA" or "GPT" or "BERT" in either the repository readme file, description, or in code saved in the repository. This group represents our treated group in the synthetic difference-in-difference set up. To identify a comparable group of repositories that we hypothesize would experience a limited change in

activity after ChatGPT release (and could therefore act as a control group), we attempted to scrape forks data from the 600 most forked public repositories with the query terms "statistics" or "data science" or "linear regression" or "time series" or "statistical analysis" in either the repository readme file, description, or in code saved in the repository. The first set of query terms is meant to identify repositories focused on building with LLMs (LLM repos) while the second is meant to identify repositories having to do with data science and statistics (DataSci repos). 24 repositories matched both sets of query terms and were conservatively assigned to the LLM repo group prior to analysis. A small subset of the queried repositories did not allow scraping of the readme files and/or forks data, yielding a final sample of 591 LLM repos and 576 DataSci repos. In total, these repositories were forked 1,645,190 times between March 7, 2008 and April 28, 2023. In Figure 19, monthly forks are plotted over time for all repositories in the sample between 2016 and 2023. Figure 20 shows monthly forks over time split by query group.

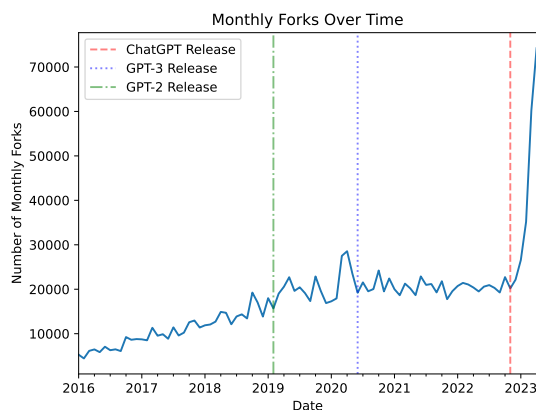


Figure 19: Number of forks per month between January 2016 and April 2023, all queried Github repositories

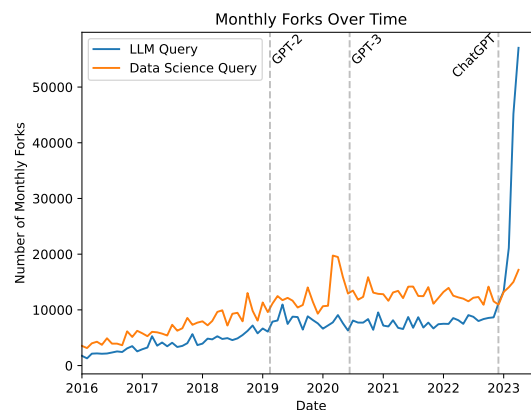


Figure 20: Number of forks per month between January 2016 and April 2023, Github repositories split by query group

These figures paint a clear picture of increased developer activity in the LLM repositories coinciding with the release of ChatGPT. To estimate the causal impact that ChatGPT release had on developer activity in LLM repos, we present results in Table 15 of three regression

specifications. Results for our main specification presented in Column 1 are that of a synthetic difference-in-difference estimator. As discussed in (81), the synthetic difference-in-difference combines aspects of the synthetic control and traditional difference-in-difference methodologies. This method is particularly useful for causal inference under quasi-experimental settings where treatment and control groups do not have parallel pre-treatment trends in aggregate (overcoming a common challenge to causal identification with traditional difference-in-difference methods). Additionally, the synthetic difference-in-difference estimator does not require that the treated unit be within a convex hull of control units (a requirement when using a traditional synthetic control estimator for causal identification).

Equation 1 represents our main synthetic difference-in-difference specification. The dataset we are using is a balanced panel with $N=1,167$ repositories and $T=122$ time periods (weeks), where the number of weekly forks for repository i in period t is denoted by Y_{it} , and exposure to the ChatGPT treatment is denoted by $W_{it} \in \{0, 1\}$. $N_{DS}=576$ (DataSci) repositories are the control units,³ while the remaining $N_{LLM}=591$ (LLM) repositories that (we contend) are differentially exposed to ChatGPT impacts after time T_{pre} .

As with a synthetic control estimator, we compute weights $\hat{\omega}_{sdid}$ that match pre-treatment trends in the number of forks of DataSci repos with those for LLM repos, e.g., $\sum_{i=1}^{N_{DS}} \hat{\omega}_{sdid} Y_{it} \approx N_{LLM}^{-1} \sum_{i=N_{DS}+1}^N Y_{it}$ for all $t = 1, \dots, T_{pre}$. We also use time weights $\hat{\lambda}_{sdid}$ that balance pre-ChatGPT weeks with post-ChatGPT weeks. Following (81), we use the synthetic difference-in-difference estimator that incorporates these weights in a two-way fixed effects regression to estimate the average causal effect of ChatGPT on weekly forks in the LLM repos (denoted by τ):

³Causal identification here relies on the assumption that these repositories were differentially exposed to ChatGPT due to their focus on code development for data science purposes rather than a focus on LLM-related coding projects. In practice, due to ChatGPT’s relevance to any kind of coding, the DataSci repos can plausibly be expected to see an increase in forking activity as well, meaning that the ATT coefficient estimates should be interpreted as a conservative estimate relative to the true ATT.

$$(\hat{\tau}_{\text{sdid}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau, \mu, \alpha, \beta} \left[\sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \hat{\omega}_{\text{sdid},i} \hat{\lambda}_{\text{sdid},t} \right] \quad (1)$$

We use the *sdid* stata package developed by (82) to estimate this average treatment effect on treated units (ATT) using the two-way fixed effects regression in equation 1 above (presented in Column 1 of table 15) as well as a traditional OLS difference-in-difference estimator (Column 2), and a synthetic control (Column 3). Graphical representations of each are presented in Figures 21 and 22.

Table 15: Impact of ChatGPT Release on Developer Activity (Weekly Forks) on Github

	(1) Synth. DiD Estimate on Weekly Forks	(2) DiD Estimate on Weekly Forks	(3) Synth. Control Estimate on Weekly Forks
LLM Repo x PostChatGPT Treatment	7.385*** (1.351)	7.795*** (1.336)	8.416*** (1.378)
N	142,374	142,374	142,374

Standard errors in parentheses

*p<0.10, **p<0.05, ***p<0.01

Table 15 shows the results of the synthetic difference in difference, estimating an average treatment effect of ChatGPT release on LLM repos of 7.385 additional forks per week, with a standard deviation of 1.351 forks per week. Since this estimator allows us to relax the parallel trends assumption, we treat it as our main specification. However, the similar point estimate in column 2 and the plotted pre-treatment trends in Figure 21 suggest that parallel trends is a reasonable identifying assumption in this instance. The synthetic control estimates in column (3) further validate the direction and magnitude of impact. We interpret these results as a strong sign that the release of ChatGPT has already driven a significant increase in developer activity, adding evidence that today’s most advanced LLMs are sparking complementary development on top of the base-models.

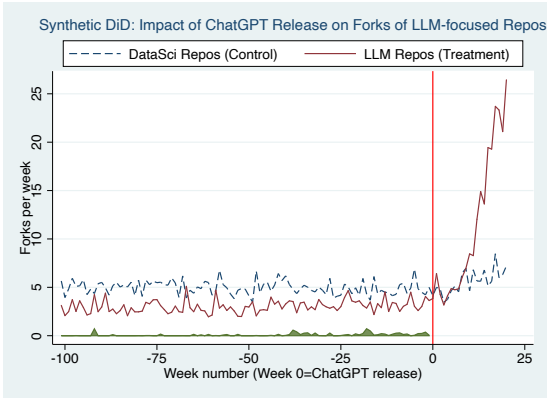


Figure 21: Pre- and post- ChatGPT release trends in weekly forks across repo groups. Weights used to average the pre-treatment time-periods across groups for the synthetic difference-in-difference are presented at the bottom of the graph. ATT estimates are presented in Columns 1 and 2 of Table 15.

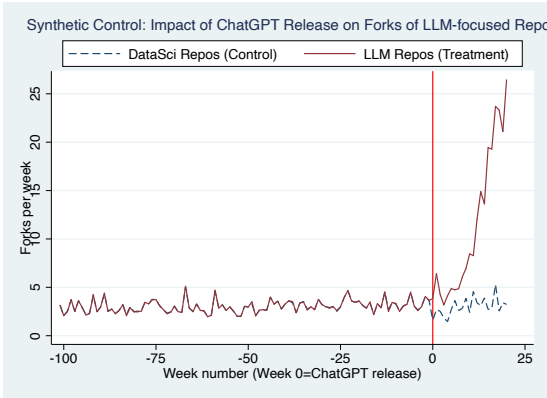
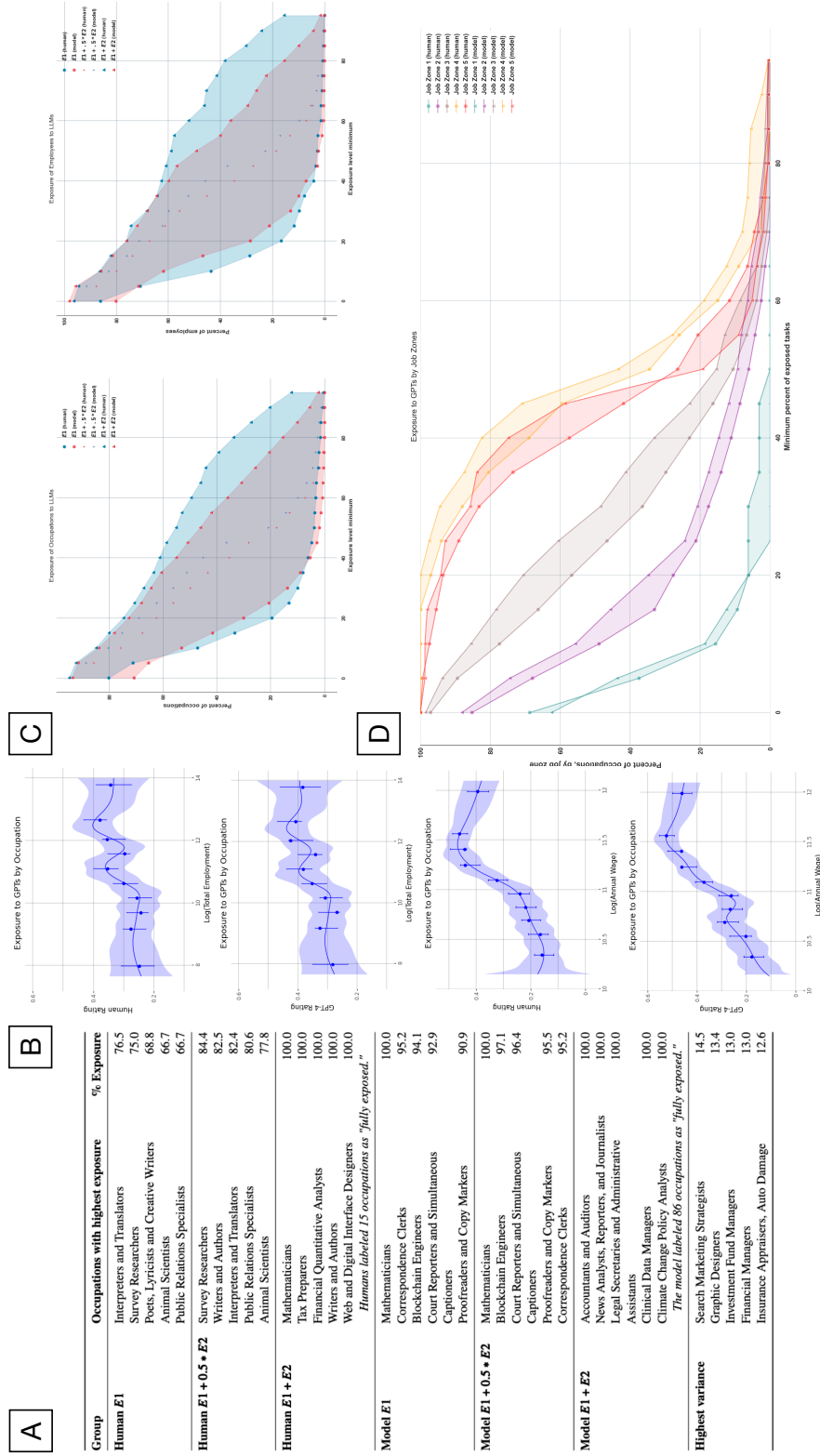


Figure 22: Synthetic Control trend line. ATT estimate is presented in Column 3 of Table 15.

941 **11 Larger Figures with Compiled Results**



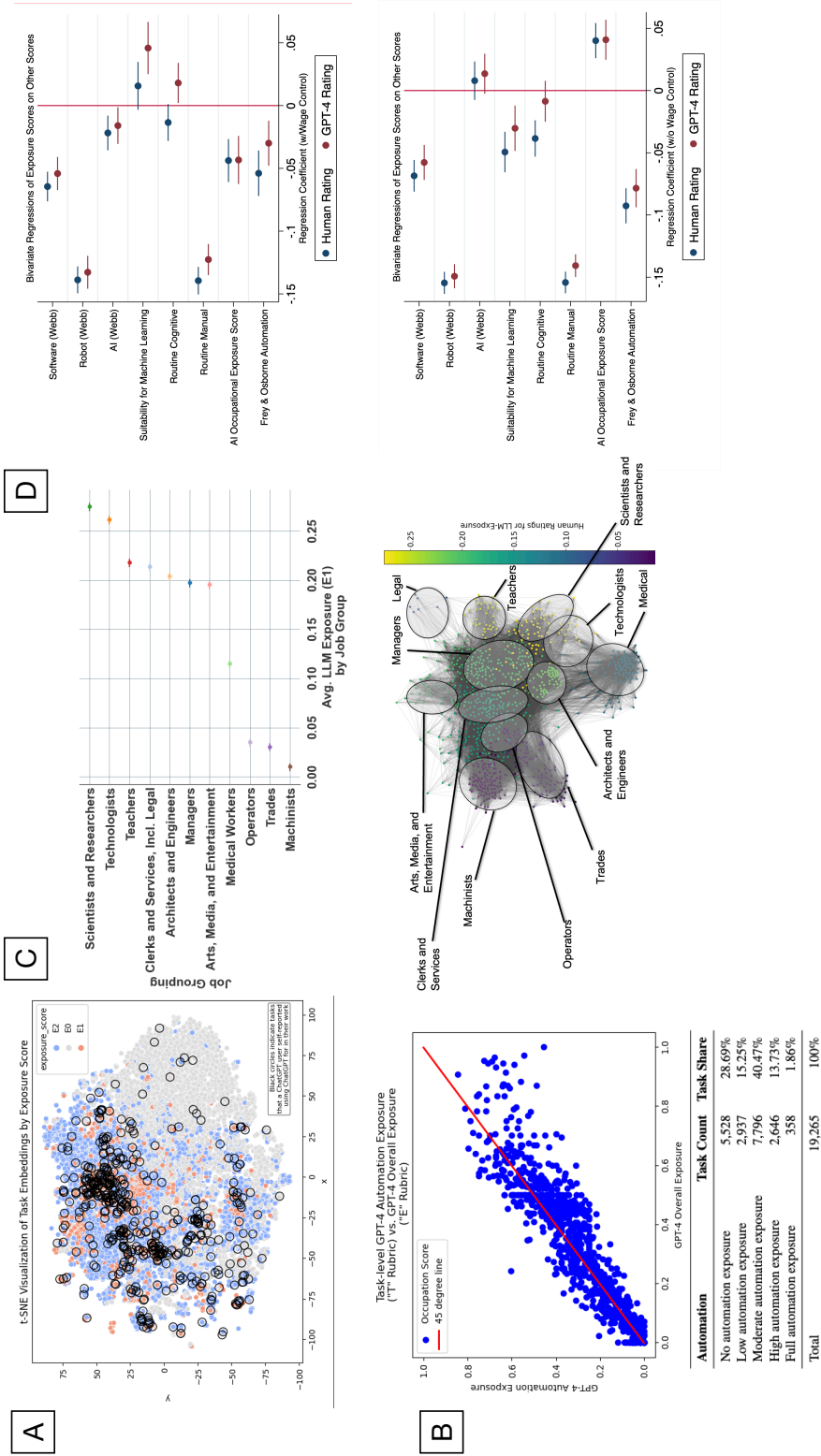


Figure 24

Figures are as described in earlier sections of the supplementary materials. A network plot connecting O*NET-SOC Codes as nodes to each other via DWAs they share in common is included in Panel C (lower graph). This graph is colored by the proportion of tasks in an occupation with a human rating of E1. Panel D also includes bivariate coefficients between the E1+0.5*E2 measure and other measures of AI or technology exposure. Specifications with and without wage controls are included.

References

1. T. Brown, *et al.*, *Advances in neural information processing systems* **33**, 1877 (2020).
2. A. Radford, *et al.*, Language models are unsupervised multitask learners, *Tech. rep.*, OpenAI (2019).
3. D. Hernandez, J. Kaplan, T. Henighan, S. McCandlish, *arXiv preprint arXiv:2102.01293* (2021).
4. J. Kaplan, *et al.*, *arXiv preprint arXiv:2001.08361* (2020).
5. OpenAI, GPT-4 Technical Report, *Tech. rep.*, OpenAI (2023).
6. L. Ouyang, *et al.*, *arXiv preprint arXiv:2203.02155* (2022).
7. Y. Bai, *et al.*, *arXiv preprint arXiv:2204.05862* (2022).
8. A. Abid, M. Farooqi, J. Zou, *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, AIES '21 (Association for Computing Machinery, New York, NY, USA, 2021), pp. 298–306.
9. P. Schramowski, C. Turan, N. Andersen, C. A. Rothkopf, K. Kersting, *Nature Machine Intelligence* **4**, 258 (2022).
10. J. A. Goldstein, *et al.*, *arXiv preprint arXiv:2301.04246* (2023).
11. OpenAI, GPT-4 System Card, *Tech. rep.*, OpenAI (2023).
12. K. Greshake, *et al.*, *arXiv preprint arXiv:2302.12173* (2023).
13. T. Schick, *et al.*, *arXiv preprint arXiv:2302.04761* (2023).

- 961 14. G. Mialon, *et al.*, *arXiv preprint arXiv:2302.07842* (2023).
- 962 15. H. Chase, LangChain (2022).
- 963 16. J. Howard, S. Ruder, *arXiv preprint arXiv:1801.06146* (2018).
- 964 17. P. Lewis, *et al.*, *Advances in Neural Information Processing Systems* **33**, 9459 (2020).
- 965 18. A. K. Singla, E. Horvitz, P. Kohli, A. Krause, *AAAI Conference on Human Computation &*
966 *Crowdsourcing* (2015).
- 967 19. D. Shahaf, E. Horvitz, *Proceedings of the AAAI Conference on Artificial Intelligence* (2010).
- 968 20. H. Khlaaf, P. Mishkin, J. Achiam, G. Krueger, M. Brundage, *arXiv preprint arXiv:2207.14157*
969 (2022).
- 970 21. L. Weidinger, *et al.*, *2022 ACM Conference on Fairness, Accountability, and Transparency,*
971 *FAccT '22* (Association for Computing Machinery, New York, NY, USA, 2022), pp. 214–229.
- 972 22. I. Solaiman, *et al.*, *arXiv preprint arXiv:1908.09203* (2019).
- 973 23. L. F. Katz, K. M. Murphy, *The Quarterly Journal of Economics* **107**, 35 (1992).
- 974 24. D. H. Autor, F. Levy, R. J. Murnane, *The Quarterly Journal of Economics* **118**, 1279 (2003).
- 975 25. D. Acemoglu, D. Autor, *Handbook of Labor Economics*, O. Ashenfelter, D. Card, eds.
976 (Elsevier, 2011), vol. 4 of *Handbook of Labor Economics*, chap. 12, pp. 1043–1171.
- 977 26. D. Acemoglu, P. Restrepo, *American Economic Review* **108**, 1488 (2018).
- 978 27. D. Acemoglu, P. Restrepo, *Journal of Economic Perspectives* **33**, 3 (2019).
- 979 28. D. H. Autor, L. F. Katz, M. S. Kearney, *American Economic Review* **96**, 189 (2006).

- 980 29. J. Van Reenen, *Labour economics* **18**, 730 (2011).
- 981 30. D. Acemoglu, P. Restrepo, *Econometrica* **90**, 1973 (2022).
- 982 31. S. Noy, W. Zhang, *Science* **381**, 187 (2023).
- 983 32. S. Peng, E. Kalliamvakou, P. Cihon, M. Demirer, *arXiv preprint arXiv:2302.06590* (2023).
- 984 33. E. Brynjolfsson, D. Li, L. R. Raymond, Generative ai at work, *Working Paper 31161*,
985 National Bureau of Economic Research (2023).
- 986 34. M. Webb, The impact of artificial intelligence on the labor market (2020). Working paper,
987 Stanford University.
- 988 35. B. Meindl, M. R. Frank, J. Mendonça, *arXiv preprint arXiv:2110.13317* (2021).
- 989 36. E. W. Felten, M. Raj, R. Seamans, *AEA Papers and Proceedings* **108**, 54 (2018).
- 990 37. E. Felten, M. Raj, R. Seamans, *arXiv preprint arXiv:2303.01157* (2023).
- 991 38. S. Tolan, *et al.*, *Journal of Artificial Intelligence Research* **71**, 191 (2021).
- 992 39. C. B. Frey, M. A. Osborne, *Technological Forecasting and Social Change* **114**, 254 (2017).
- 993 40. M. Arntz, T. Gregory, U. Zierahn, *Economics Letters* **159**, 157 (2017).
- 994 41. K. Grace, J. Salvatier, A. Dafoe, B. Zhang, O. Evans, *Journal of Artificial Intelligence*
995 *Research* **62**, 729 (2018).
- 996 42. E. Brynjolfsson, T. Mitchell, D. Rock, *AEA Papers and Proceedings* **108**, 43 (2018).
- 997 43. E. Brynjolfsson, M. R. Frank, T. Mitchell, I. Rahwan, D. Rock, *Working Paper* (2023).

- 998 44. D. Autor, C. Chin, A. M. Salomons, B. Seegmiller, New frontiers: The origins and content
999 of new work, 1940–2018, *Working Paper 30389*, National Bureau of Economic Research
1000 (2022).
- 1001 45. M. R. Frank, *et al.*, *Proceedings of the National Academy of Sciences* **116**, 6531 (2019).
- 1002 46. R. Bommasani, *et al.*, *arXiv preprint arXiv:2108.07258* (2021).
- 1003 47. A. Korinek, Language models and cognitive automation for economic research, *Working*
1004 *Paper 30957*, National Bureau of Economic Research (2023).
- 1005 48. E. R. Mollick, L. Mollick, *SSRN Electronic Journal* (2023).
- 1006 49. T. F. Bresnahan, M. Trajtenberg, *Journal of Econometrics* **65**, 83 (1995).
- 1007 50. R. G. Lipsey, K. I. Carlaw, C. T. Bekar, *Economic Transformations: General Purpose*
1008 *Technologies and Long Term Economic Growth* (Oup Oxford, 2005).
- 1009 51. J. Bessen, *The economics of artificial intelligence: an agenda* (University of Chicago Press,
1010 2018), pp. 291–307.
- 1011 52. A. Korinek, J. E. Stiglitz, *The economics of artificial intelligence: an agenda* (University of
1012 Chicago Press, 2018), pp. 349–390.
- 1013 53. D. Acemoglu, D. Autor, J. Hazell, P. Restrepo, Ai and jobs: Evidence from online vacancies,
1014 *Working Paper 28257*, National Bureau of Economic Research (2020).
- 1015 54. S. G. Benzell, L. J. Kotlikoff, G. LaGarda, V. Y. Ye, Simulating endogenous global automation,
1016 *Working Paper 29220*, National Bureau of Economic Research (2021).
- 1017 55. T. Bresnahan, S. Greenstein, D. Brownstone, K. Flamm, *Brookings Papers on Economic*
1018 *Activity. Microeconomics* **1996**, 1 (1996).

- 1019 56. T. F. Bresnahan, E. Brynjolfsson, L. M. Hitt, *The Quarterly Journal of Economics* **117**, 339
1020 (2002).
- 1021 57. J. Dixon, B. Hong, L. Wu, *Management Science* **67**, 5586 (2021).
- 1022 58. P. A. David, *American Economic Review* **80**, 355 (1990).
- 1023 59. T. F. Bresnahan, *The economic journal* **109**, 390 (1999).
- 1024 60. C. B. Frey, *The Technology Trap* (Princeton University Press, 2019).
- 1025 61. E. Brynjolfsson, D. Rock, C. Syverson, *American Economic Journal: Macroeconomics* **13**,
1026 333 (2021).
- 1027 62. J. J. Feigenbaum, D. P. Gross, Organizational frictions and increasing returns to automation:
1028 Lessons from at&t in the twentieth century, *Working Paper 29580*, National Bureau of
1029 Economic Research (2021).
- 1030 63. T. Bresnahan, *Artificial Intelligence Technologies and Aggregate Growth Prospects* (Cam-
1031 bridge University Press, 2021), pp. 132–170.
- 1032 64. A. K. Agrawal, J. S. Gans, A. Goldfarb, AI adoption and system-wide change, *Working*
1033 *Paper 28811*, National Bureau of Economic Research (2021).
- 1034 65. A. Goldfarb, B. Taska, F. Teodoridis, *Research Policy* **52**, 104653 (2023).
- 1035 66. D. Rock, *SSRN Electronic Journal* (2019).
- 1036 67. T. Babina, A. Fedyk, A. He, J. Hodson, *Firm Growth, and Product Innovation (November 9,*
1037 *2021)* (2021).

- 1038 68. N. Zolas, *et al.*, Advanced technologies adoption and use by us firms: Evidence from the
1039 annual business survey, *Working Paper 28290*, National Bureau of Economic Research
1040 (2021).
- 1041 69. I. M. Cockburn, R. Henderson, S. Stern, *The economics of artificial intelligence: An agenda*
1042 (University of Chicago Press, 2018), pp. 115–146.
- 1043 70. Z. Cheng, D. Lee, P. Tambe, *SSRN Electronic Journal* (2022).
- 1044 71. T. Besiroglu, N. Emery-Xu, N. Thompson, *arXiv preprint arXiv:2212.08198* (2022).
- 1045 72. M.-H. Huang, R. T. Rust, *Journal of service research* **21**, 155 (2018).
- 1046 73. E. Brynjolfsson, T. Mitchell, *Science* **358**, 1530 (2017).
- 1047 74. O*NET, O*NET 27.2 Database (2023).
- 1048 75. BLS, Employment by detailed occupation (2022).
- 1049 76. BLS, Occupational Outlook Handbook A-Z Index (2023).
- 1050 77. BLS, Demographic Characteristics (CPS) (2023).
- 1051 78. D. Acemoglu, D. Autor, *Handbook of labor economics* (Elsevier, 2011), vol. 4, pp. 1043–
1052 1171.
- 1053 79. L. Kogan, D. Papanikolaou, L. D. W. Schmidt, B. Seegmiller, Technology, vintage-specific
1054 human capital, and labor displacement: Evidence from linking patents with occupations,
1055 *Working Paper 29552*, National Bureau of Economic Research (2021).
- 1056 80. E. Davis, S. Aaronson, *arXiv preprint arXiv:2308.05713* (2023).

- 1057 81. D. Arkhangelsky, S. Athey, D. A. Hirshberg, G. W. Imbens, S. Wager, *American Economic*
1058 *Review* **111**, 4088 (2021).
- 1059 82. D. Clarke, D. Paila, S. Athey, G. Imbens, Synthetic difference-in-differences estimation,
1060 *Tech. rep.*, IZA Discussion Paper (2023).

1061 **LLM assistance statement**

1062 In addition to using GPT-4 to rate task exposure according the rubrics discussed previously
1063 in the Supplementary Materials, GPT-4 and ChatGPT were also used for writing, coding, and
1064 formatting assistance in this project. After using these tools, the authors reviewed and edited text
1065 and code as needed and take full responsibility for the content of the publication.

1066 **11.1 Sample Prompts for GPT-4 Assistance**

- 1067 • Can you give me a citation for this book:
1068 The Work of the Future Building Better Jobs in an Age of Intelligent Machines By David
1069 Autor, David A. Mindell and Elisabeth B. Reynolds
1070 that is formatted for Science magazine? let me know if you need more information about
1071 the book.
- 1072 • Can you please edit references so they are in latex format?
- 1073 • I need to place a figure anchored in a certain spot in my Latex document. Can you give me
1074 some sample latex code to show me how to do that?
- 1075 • (code) Given the sample df, generate unique id that is a concatenation of onetsoccode and
1076 taskid. double check that it is unique.

- 1077 • (code) Given the sample df, provide some starter code to plot the correlation between the
1078 importance-weighted and core-weighted human ratings, include axis labels and title.
- 1079 • Please remove a sentence from the following paragraph without changing the content
1080 described: paragraph inserted here.
- 1081 • (code) I need help creating a Sankey diagram to show the relationship between three sets
1082 of dummy variables. What is a good python package to use for that?
- 1083 • (code) Please give me some sample python code to create a Sankey Diagram with white
1084 nodes and custom link colors.

1085 **12 Publicly Accessible Code and Data**

1086 All code and data materials are available at the following link: [https://github.com/openai/GPTs-](https://github.com/openai/GPTs-are-GPTs)
1087 are-GPTs